ENTERPRISE LEGAL AI SUITE • AUTOMATED QUALITY ASSURANCE

LegalSight — Automated E2E Selenium Test Suite & Visual Storyboard Report

High-fidelity visual verification, terminal telemetry, and structural validation across 10 modular end-to-end test suites.

TEST SUITES PASSED	TOTAL SNAPSHOTS	PASS RATE	VIEWPORT TARGET
10 / 10	37	100.0%	1440x900

SYSTEM ARCHITECTURE & TECHNICAL DEFINITIONS

Selenium WebDriver 4.51: An enterprise browser automation framework that drives real Chrome/Edge engines via the W3C WebDriver Protocol. It executes native user interactions (keystrokes, mouse events, navigation, modal states, tab switching) against live DOM trees without mocking, ensuring absolute fidelity of production workflows.

LegalSight Storyboarder Engine: A custom orchestration and visual capture layer (`storyboarder.py`) engineered specifically for legal technology testing. It records discrete operational steps with high-contrast terminal logging (safe for Windows terminals), captures calibrated 1440×900 viewport screenshots, injects timing telemetry, and outputs interactive dark-themed HTML galleries and standalone PDF reports.

Modular Architecture: To support strict test isolation and independent debugging, all 10 scenarios are decoupled into dedicated test scripts (`test_01` through `test_10`) that can be executed in isolation or unified via `run_storyboard.py`.

#	TEST ID	SUITE NAME	STEPS	STATUS
01	test_01_auth_flow	User Authentication & RBAC Access Control	6 Steps	**PASSED**
02	test_02_dashboard_ingest	Contract Ingestion Hub & Playbook Calibration	4 Steps	**PASSED**
03	test_03_reviewer_analysis	Contract Reviewer, Redlines & Certified Sign-Off	6 Steps	**PASSED**
04	test_04_datasets_models	Legal Datasets, Benchmark Models & Neural Evaluator	4 Steps	**PASSED**
05	test_05_reports_deadlines	Portfolio Liability Matrix & Renewal Deadlines Hub	3 Steps	**PASSED**

#	TEST ID	SUITE NAME	STEPS	STATUS
06	test_06_semantic_search	Semantic Legal Search & Precedent Discovery	4 Steps	**PASSED**
07	test_07_smart_contracts	Smart Contract Studio & EVM Solidity Synthesis	5 Steps	**PASSED**
08	test_08_governance_admin	Enterprise Governance, RBAC & Model Diagnostics	5 Steps	**PASSED**
09	test_09_audit_ledger	Immutable Compliance Audit Ledger & Security Log	3 Steps	**PASSED**
10	test_10_end_to_end_journey	Complete Enterprise Legal Counsel End-to-End Lifecycle	9 Steps	**PASSED**

TEST SUITE 01 OF 10

User Authentication & Role-Based Access Control (RBAC)

✓ PASSED (100%)

`test_01_auth_flow.py`

LEGAL & ARCHITECTURAL DEFINITION:

Validates the enterprise identity lifecycle, JWT bearer authentication, and strict role separation between General Counsel, Senior Analyst, and Guest tiers. Ensures that redline modification privileges, risk sensitivity sliders, and corporate governance controls remain strictly segregated by cryptographically verified session tokens.

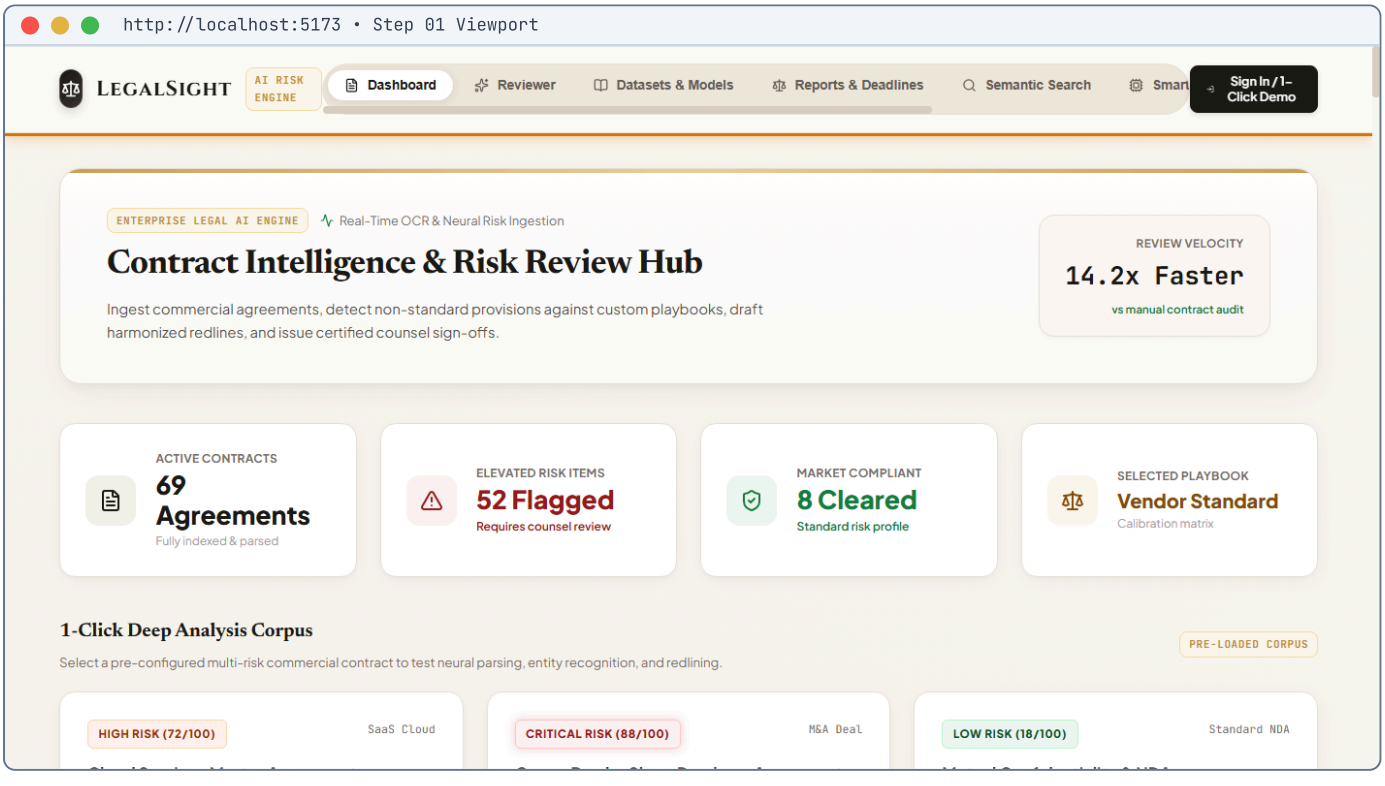
```
PowerShell (user: aarya @ laptop-t4esujeo) — D:\PROJECTS\LegalSight

+============================================================================+
|  LEGALSIGHT STORYBOARDER: TEST_01_AUTH_FLOW
|  User Authentication & Role-Based Access Control (RBAC)
+============================================================================+
  [*] Initializing CHROME browser (headless=True, viewport=1440x900)...
  [+] Browser ready. Target URL: http://localhost:5173
  [STEP 01] Landing Page & Guest Header
            > Opened LegalSight homepage in guest mode; verifying navigation links and Sign In button.
            * Snapshot: tests\e2e\screenshots\test_01_auth_flow\01_landing_page_&_guest_header.png (0.93s)
  [STEP 02] Authentication Modal Opened
            > Displayed authentication modal featuring 1-Click Instant Demo Credentials for Analyst & Admin.
            * Snapshot: tests\e2e\screenshots\test_01_auth_flow\02_authentication_modal_opened.png (0.73s)
  [STEP 03] Analyst Authenticated & Session Active
            > Successfully authenticated as Analyst. Header displays user profile badge: 'analyst (Analyst)'.
            * Snapshot: tests\e2e\screenshots\test_01_auth_flow\03_analyst_authenticated_&_sessio.png (0.45s)
  [STEP 04] User Signed Out
```

VISUAL STORYBOARD BROWSER SNAPSHOTS (6 CAPTURED)

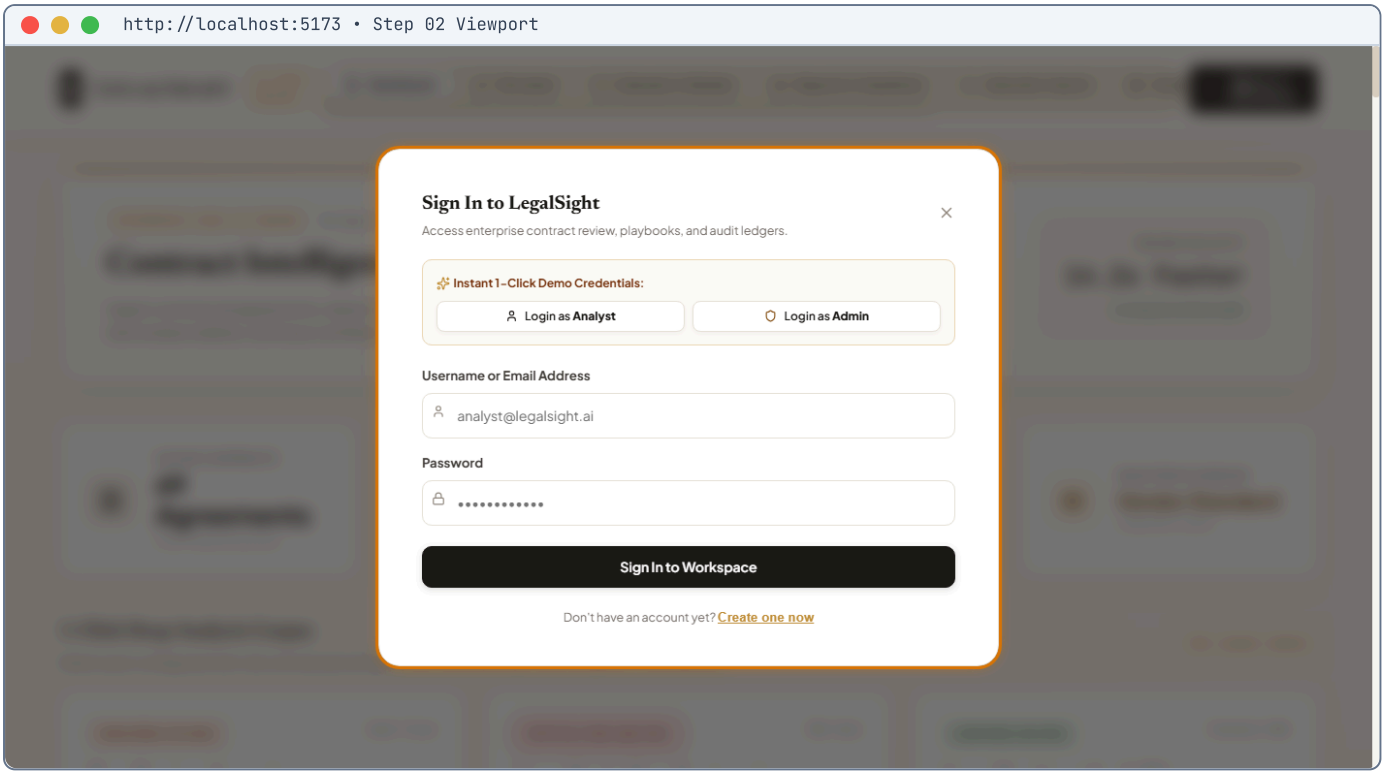
STEP 01 Landing Page & Guest Header

Opened LegalSight homepage in guest mode; verifying navigation links and Sign In button.



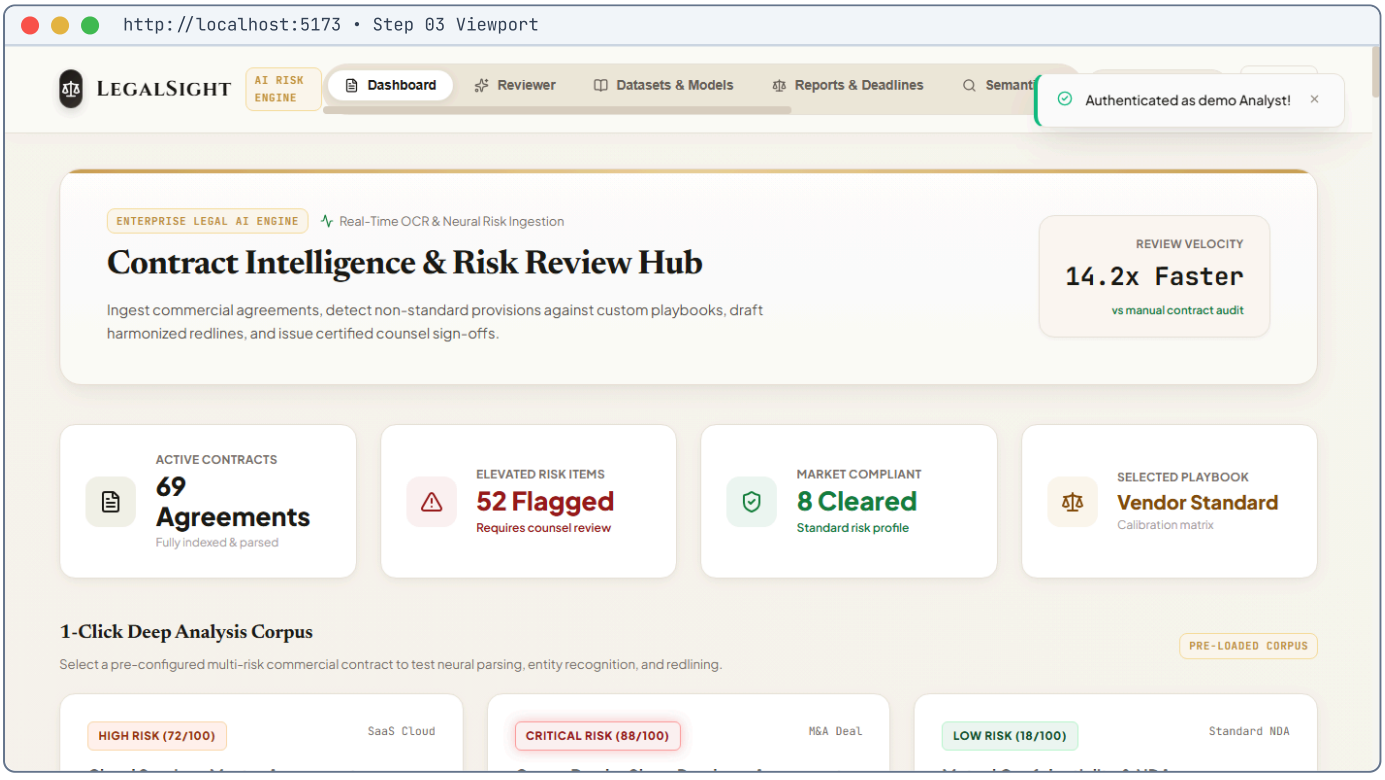
STEP 02 Authentication Modal Opened

Displayed authentication modal featuring 1-Click Instant Demo Credentials for Analyst & Admin.



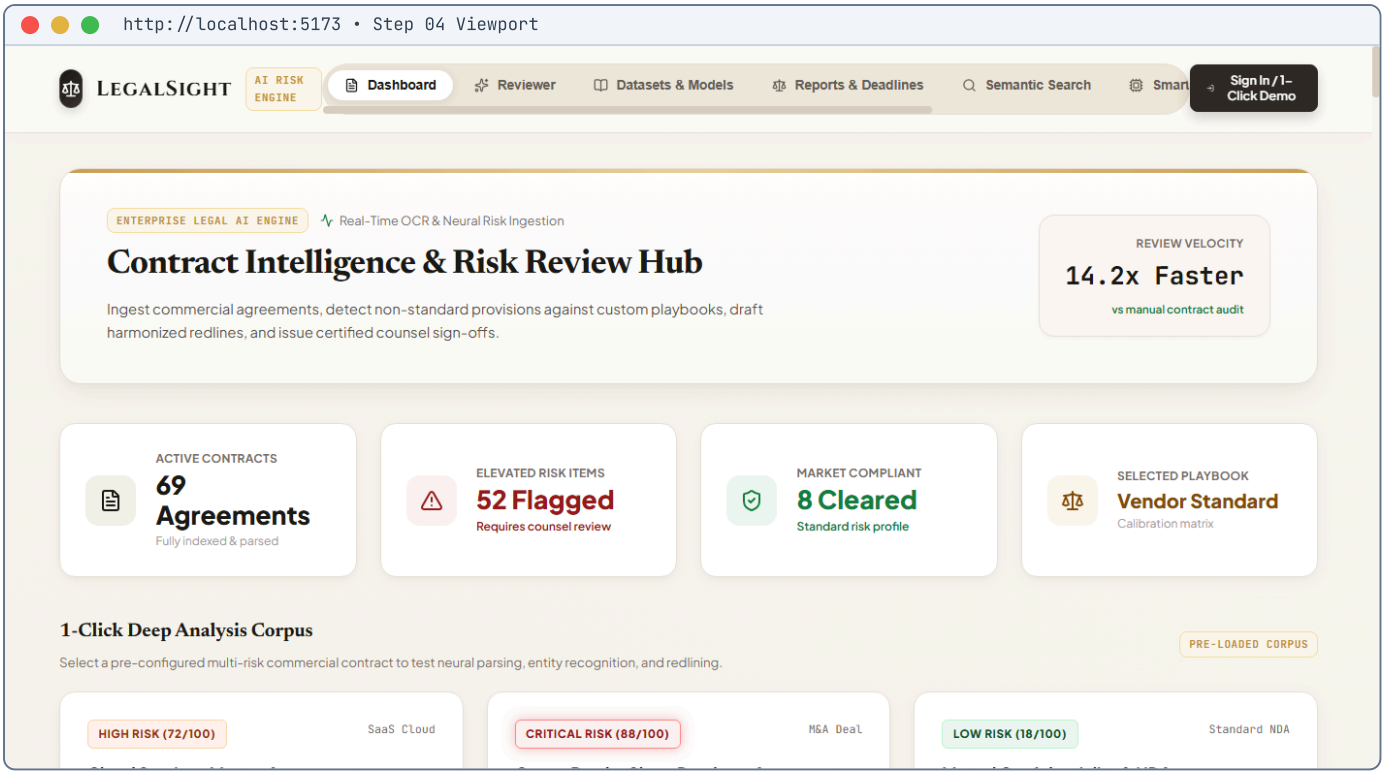
STEP 03 Analyst Authenticated & Session Active

Successfully authenticated as Analyst. Header displays user profile badge: 'analyst (Analyst)'.



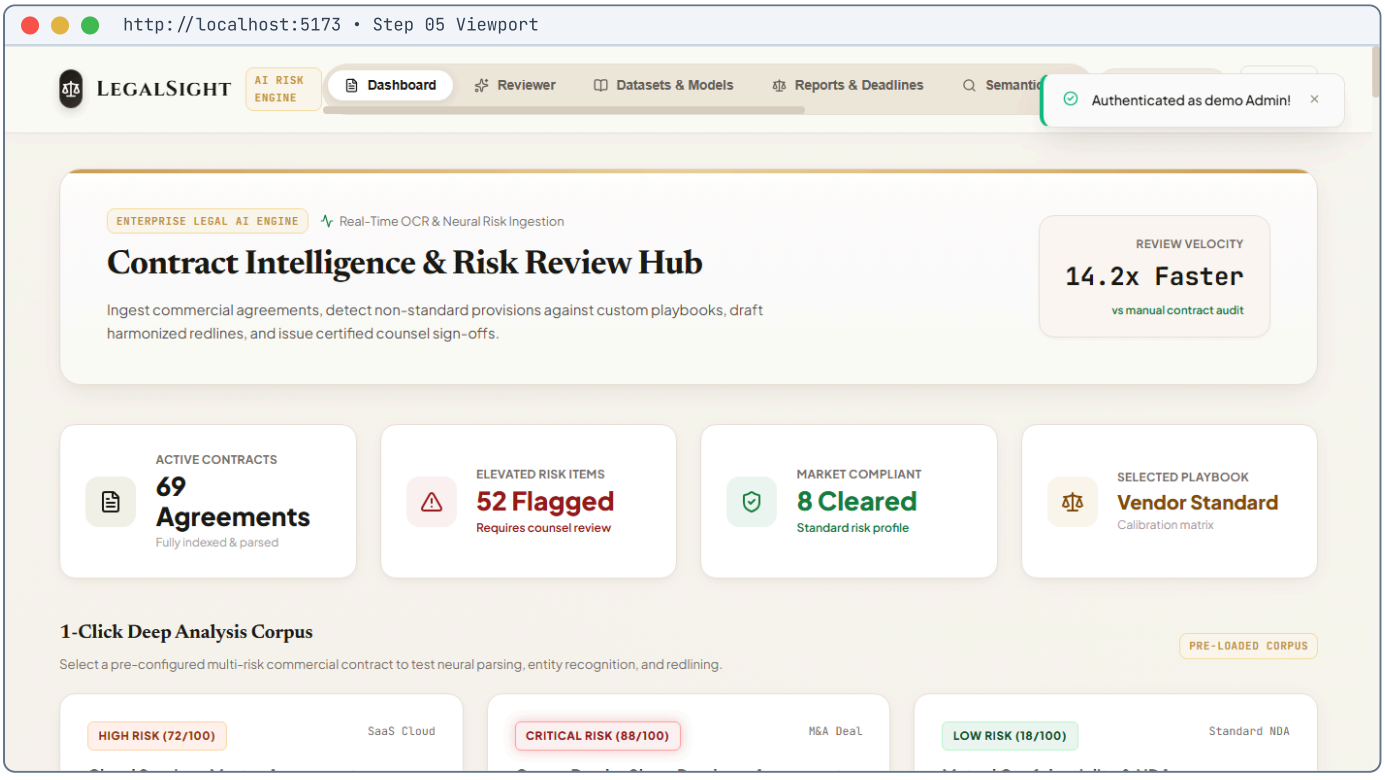
STEP 04 User Signed Out

Session cleared, authentication token invalidated, and workspace returned to guest mode.



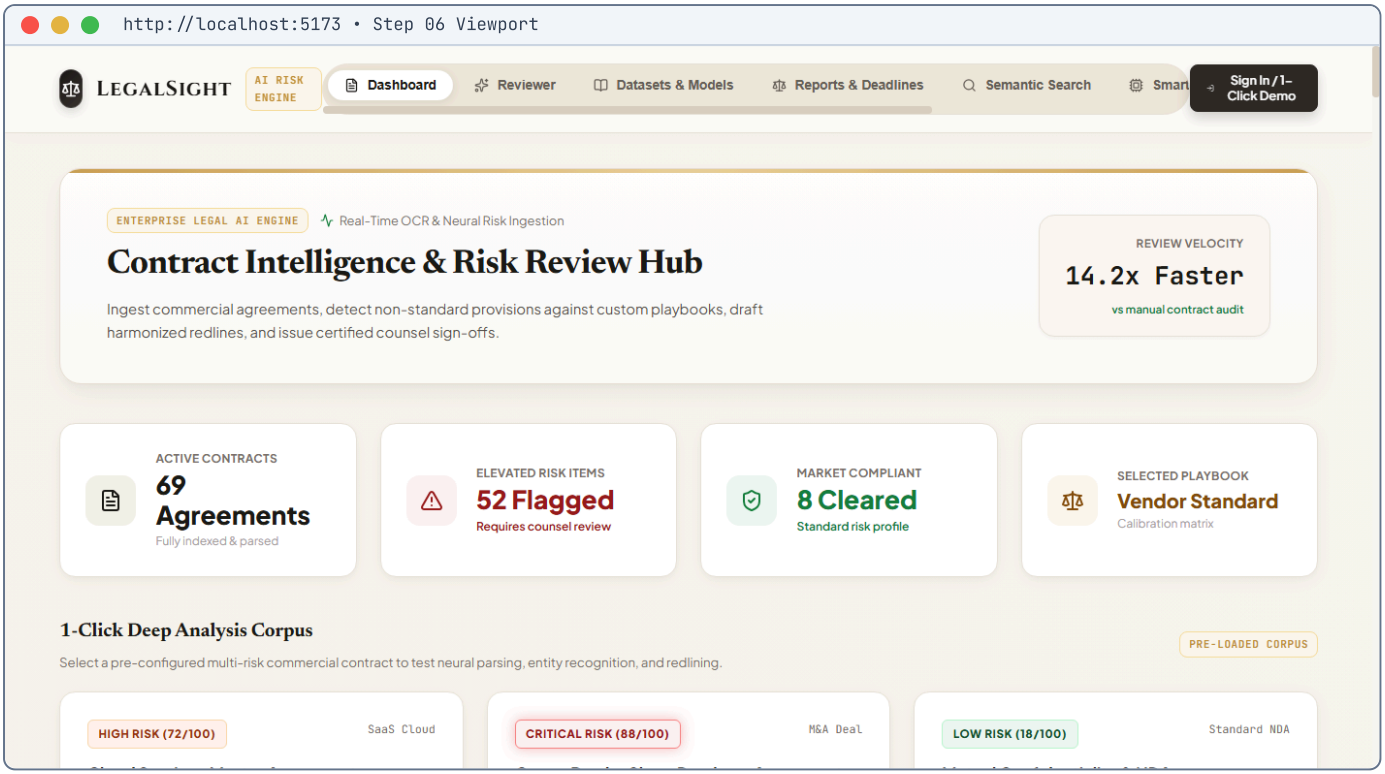
STEP 05 Admin Authenticated with Governance Rights

Successfully logged in as Admin user. Elevated governance controls and calibration consoles unlocked.



STEP 06 **Clean Workspace Reset**

Verified complete authentication lifecycle test case and reset workspace state.



TEST SUITE 02 OF 10

Contract Ingestion Hub & Playbook Calibration

✓ PASSED (100%)

`test_02_dashboard_ingest.py`

LEGAL & ARCHITECTURAL DEFINITION:

Validates the multi-format document ingestion pipeline (PDF, DOCX, OCR text) and risk calibration playbook switching. Tests pre-loaded commercial corpora including SaaS Master Services Agreements, Cross-Border Share Purchase Agreements, and Mutual NDAs, verifying automated clause extraction and seamless transition into the neural reviewer workspace.

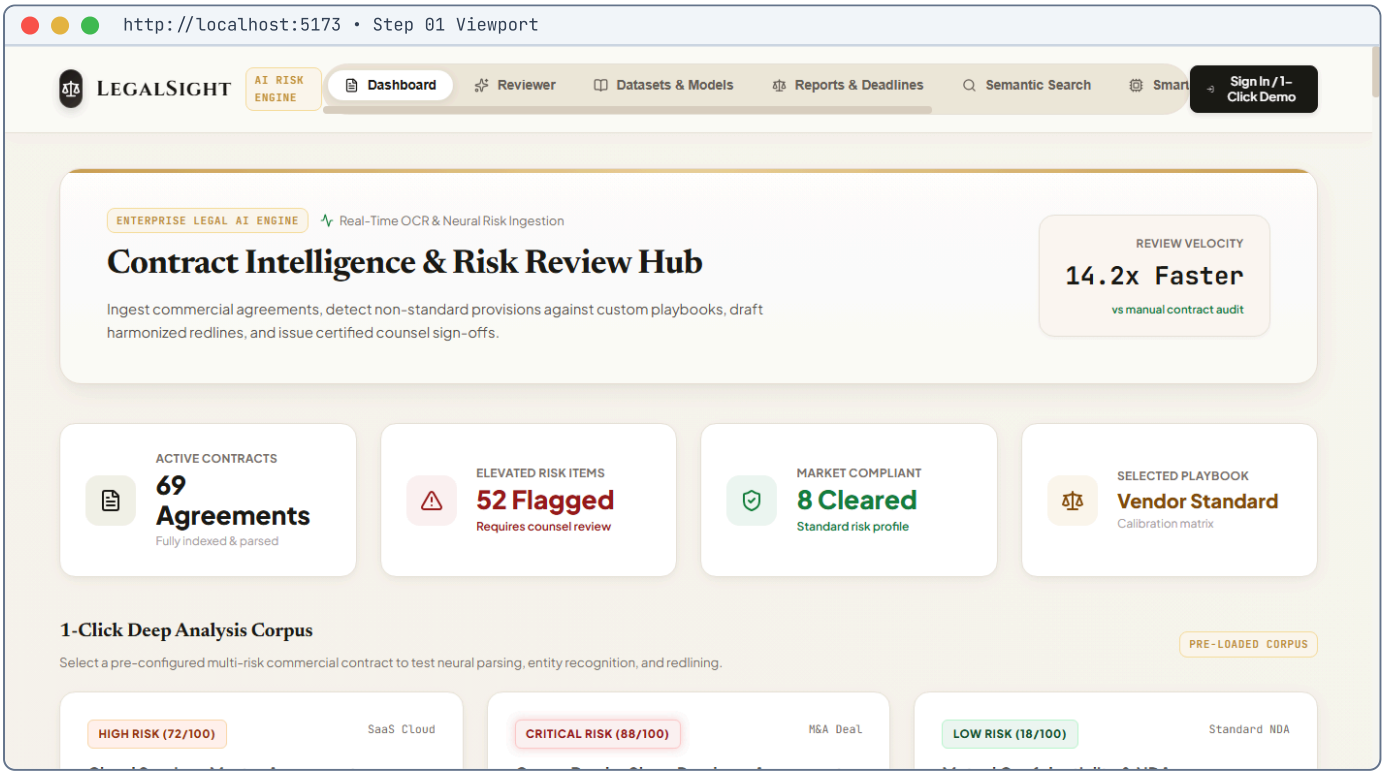
```
PowerShell (user: aarya @ laptop-t4esujeo) — D:\PROJECTS\LegalSight

+============================================================================+
|  LEGALSIGHT STORYBOARDER: TEST_02_DASHBOARD_INGEST
|  Contract Ingestion Hub & Playbook Calibration
+============================================================================+
  [*] Initializing CHROME browser (headless=True, viewport=1440x900)...
  [+] Browser ready. Target URL: http://localhost:5173
  [STEP 01] Contract Intelligence & Review Hub Overview
            > Loaded enterprise ingestion dashboard displaying active contract metrics, risk indicators, and velocity KPIs.
            * Snapshot: tests\e2e\screenshots\test_02_dashboard_ingest\01_contract_intelligence_&_review.png (0.71s)
  [STEP 02] Pre-Loaded Multi-Risk Commercial Corpus
            > Verified 1-Click ingestion options: Cloud SaaS Master Agreement (High Risk), Cross-Border Share Purchase Agreement
(Critical Risk), Mutual NDA (Low Risk).
            * Snapshot: tests\e2e\screenshots\test_02_dashboard_ingest\02_pre-loaded_multi-risk_commerci.png (0.61s)
  [STEP 03] Switched Playbook Calibration: M&A Aggressive
            > Activated M&A Aggressive calibration matrix; lowered tolerance threshold to 40 for indemnification, super-caps, and
IP covenants.
```

VISUAL STORYBOARD BROWSER SNAPSHOTS (4 CAPTURED)

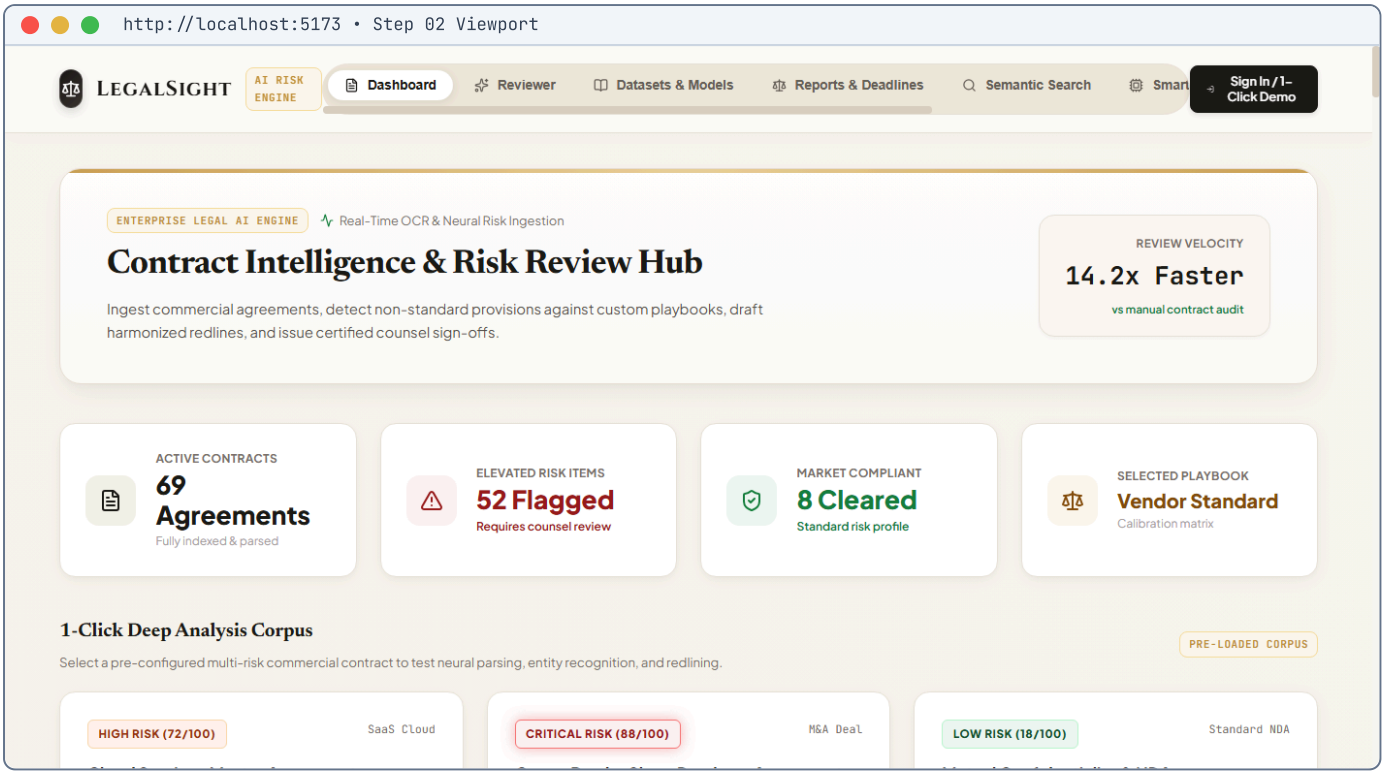
STEP 01 Contract Intelligence & Review Hub Overview

Loaded enterprise ingestion dashboard displaying active contract metrics, risk indicators, and velocity KPIs.



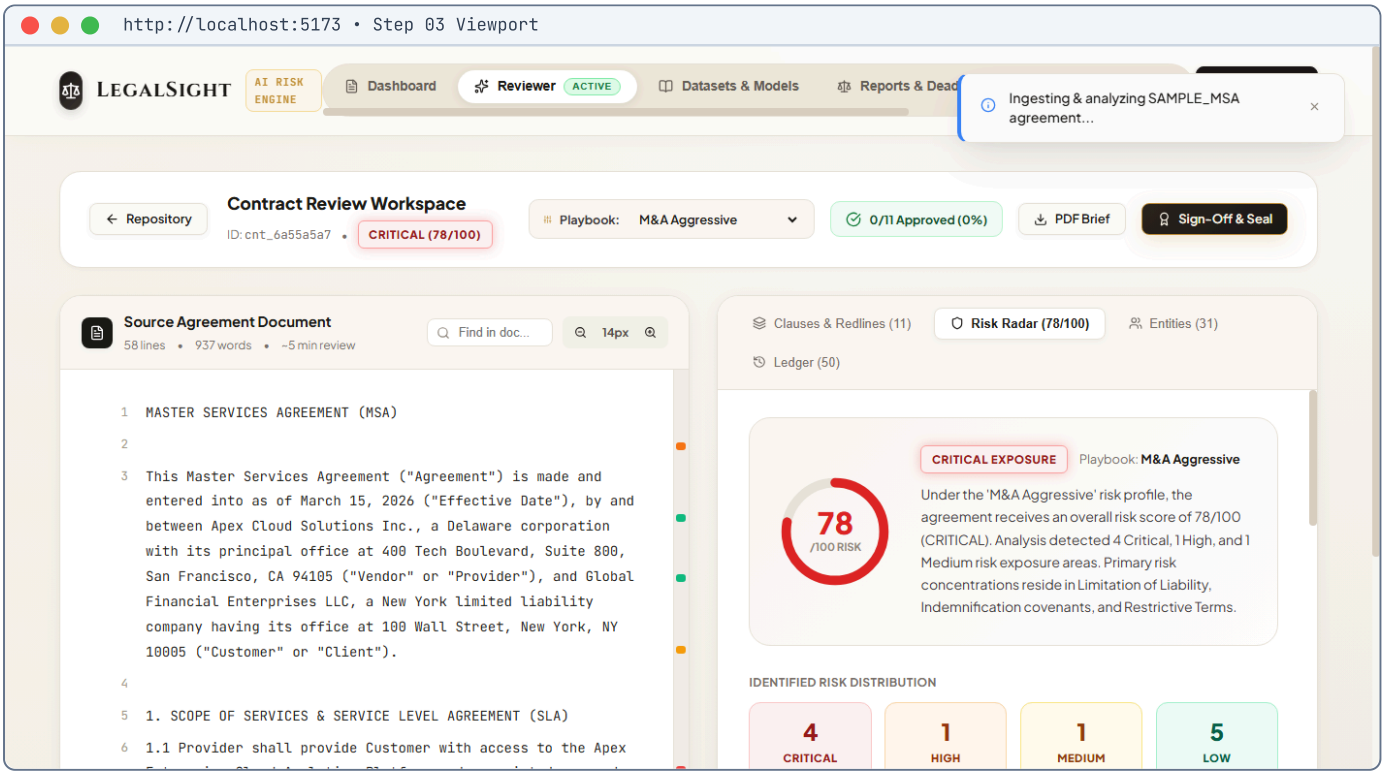
STEP 02 Pre-Loaded Multi-Risk Commercial Corpus

Verified 1-Click ingestion options: Cloud SaaS Master Agreement (High Risk), Cross-Border Share Purchase Agreement (Critical Risk), Mutual NDA (Low Risk).



STEP 03 Switched Playbook Calibration: M&A Aggressive

Activated M&A Aggressive calibration matrix; lowered tolerance threshold to 40 for indemnification, super-caps, and IP covenants.



STEP 04 **Contract Ingested & Transitioned to Reviewer**

Ingestion pipeline parsed SaaS Master Services Agreement; extracted clauses and initialized neural risk analysis workspace.

TEST SUITE 03 OF 10

Contract Reviewer, Redline Diff & Legal Seal Sign-Off

✓ PASSED (100%)

`test_03_reviewer_analysis.py`

LEGAL & ARCHITECTURAL DEFINITION:

Validates the interactive split-pane legal reviewer: raw contract rendering on the left and AI-detected clause cards on the right. Tests the General Counsel executive synthesis, Named Entity Recognition (NER), side-by-side redline comparison of non-standard indemnities, and execution of the digital attestation modal minting an immutable Gold Holographic Legal Seal.

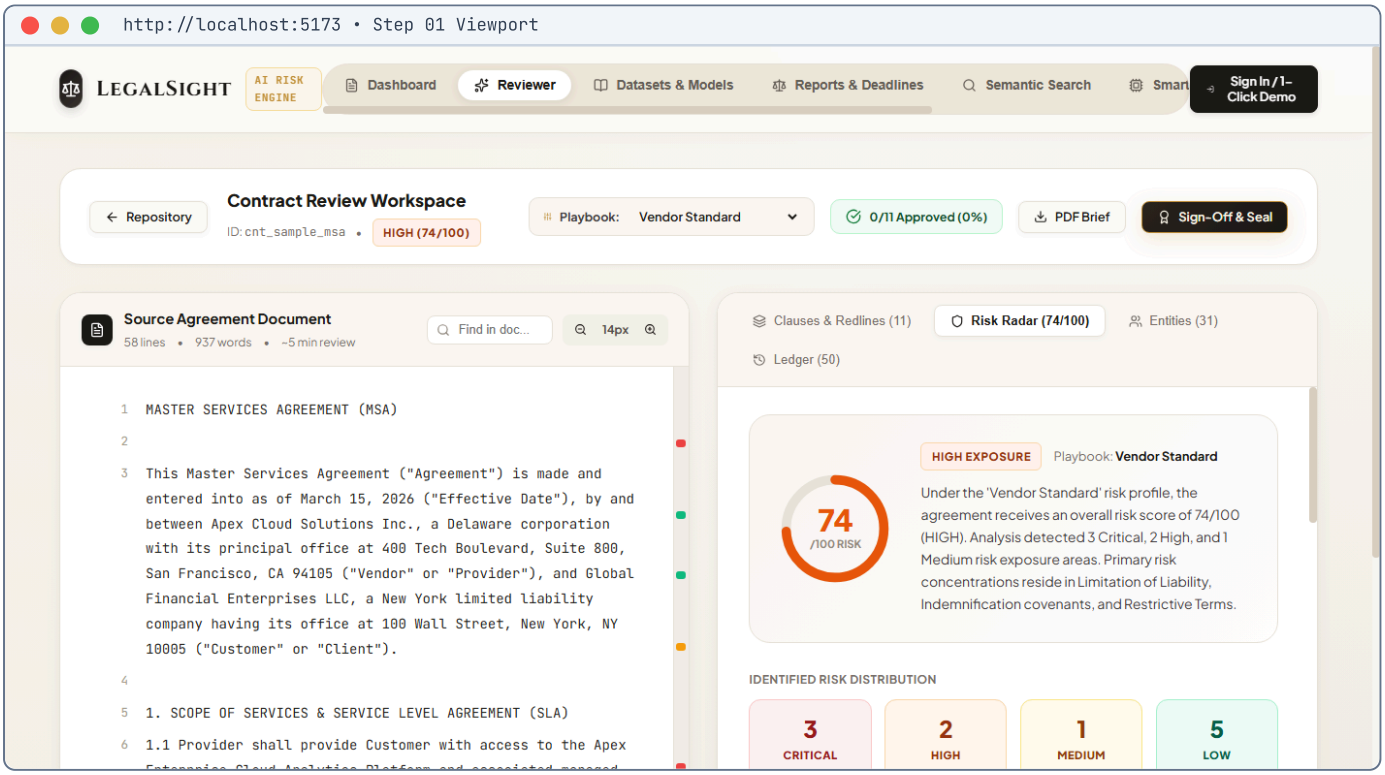
```
PowerShell (user: aarya @ laptop-t4esujeo) — D:\PROJECTS\LegalSight

+============================================================================+
|  LEGALSIGHT STORYBOARDER: TEST_03_REVIEWER_ANALYSIS
|  Contract Reviewer, Redline Diff & Legal Seal Sign-Off
+============================================================================+
  [*] Initializing CHROME browser (headless=True, viewport=1440x900)...
  [+] Browser ready. Target URL: http://localhost:5173
  [STEP 01] Split-Pane Legal Reviewer Workspace
            > Loaded contract reviewer displaying full source text on left and neural clause cards with risk radar on right.
            * Snapshot: tests\e2e\screenshots\test_03_reviewer_analysis\01_split-pane_legal_reviewer_work.png (0.97s)
  [STEP 02] General Counsel Synthesis & Risk Radar
            > Synthesized contract risk overview, key exposure warnings, and compliance scores.
            * Snapshot: tests\e2e\screenshots\test_03_reviewer_analysis\02_general_counsel_synthesis_&_ri.png (0.58s)
  [STEP 03] Extracted Entities & Legal Metadata
            > Extracted contracting parties, governing jurisdiction, liability ceilings, and termination provisions.
            * Snapshot: tests\e2e\screenshots\test_03_reviewer_analysis\03_extracted_entities_&_legal_met.png (0.56s)
  [STEP 04] Clause Cards & Proposed Redline Alternatives
```

VISUAL STORYBOARD BROWSER SNAPSHOTS (6 CAPTURED)

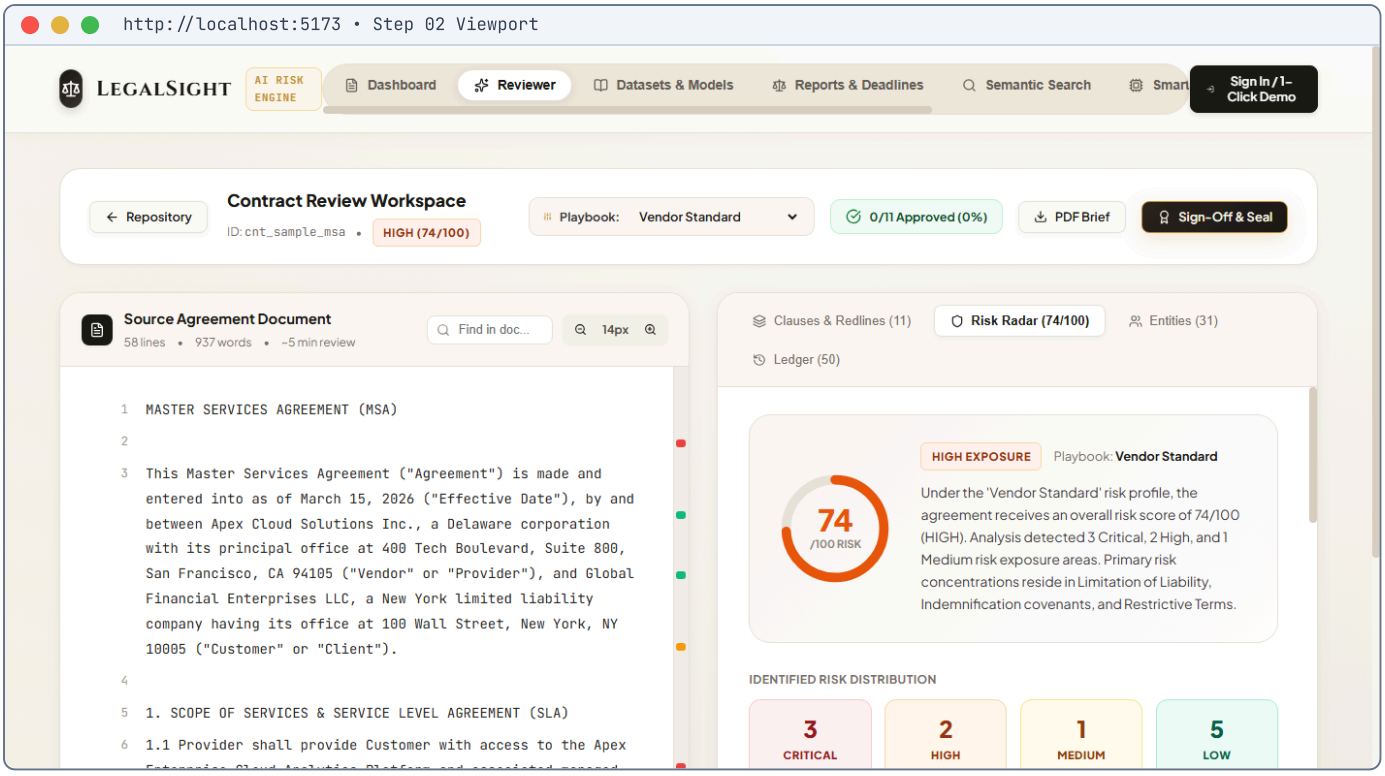
STEP 01 Split-Pane Legal Reviewer Workspace

Loaded contract reviewer displaying full source text on left and neural clause cards with risk radar on right.



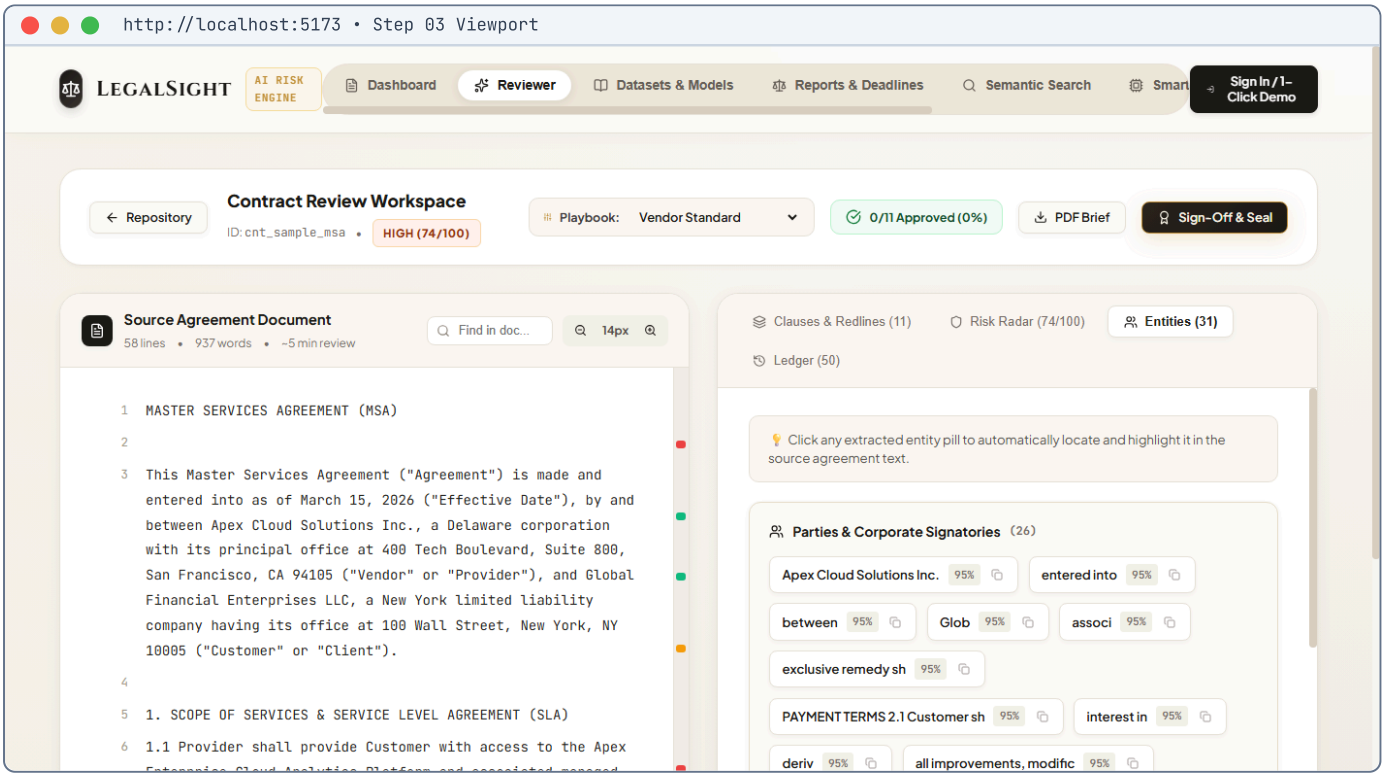
STEP 02 General Counsel Synthesis & Risk Radar

Synthesized contract risk overview, key exposure warnings, and compliance scores.



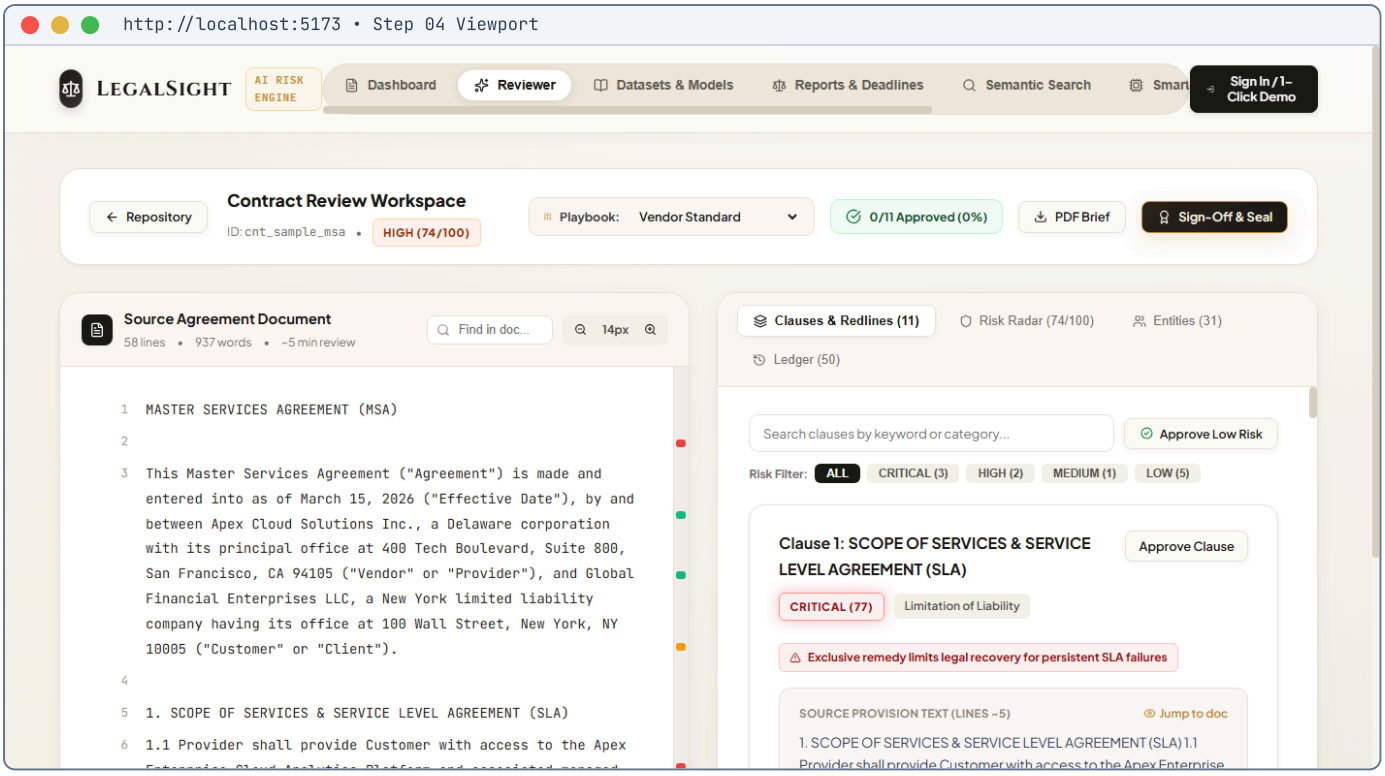
STEP 03 Extracted Entities & Legal Metadata

Extracted contracting parties, governing jurisdiction, liability ceilings, and termination provisions.



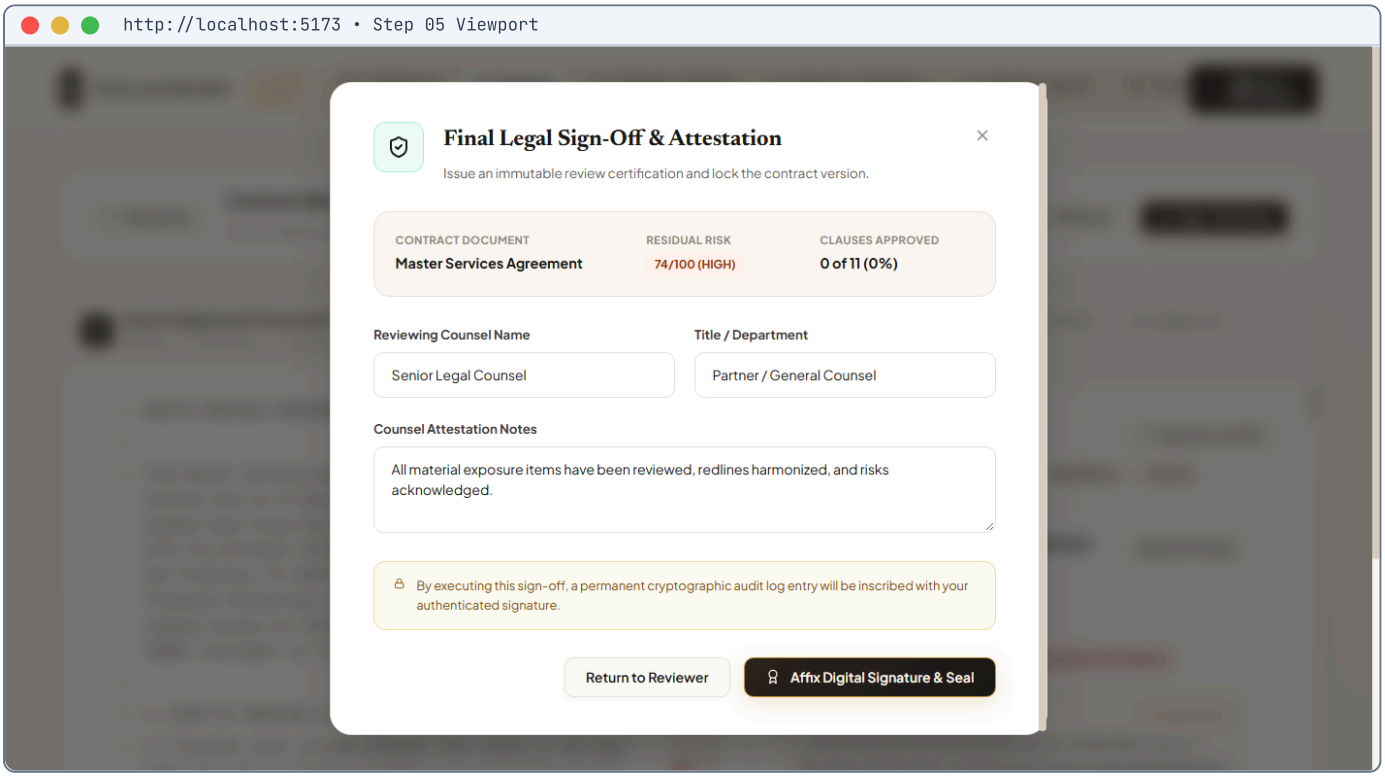
STEP 04 Clause Cards & Proposed Redline Alternatives

Viewed side-by-side redlines comparing original problematic clause against market-standard harmonized text.



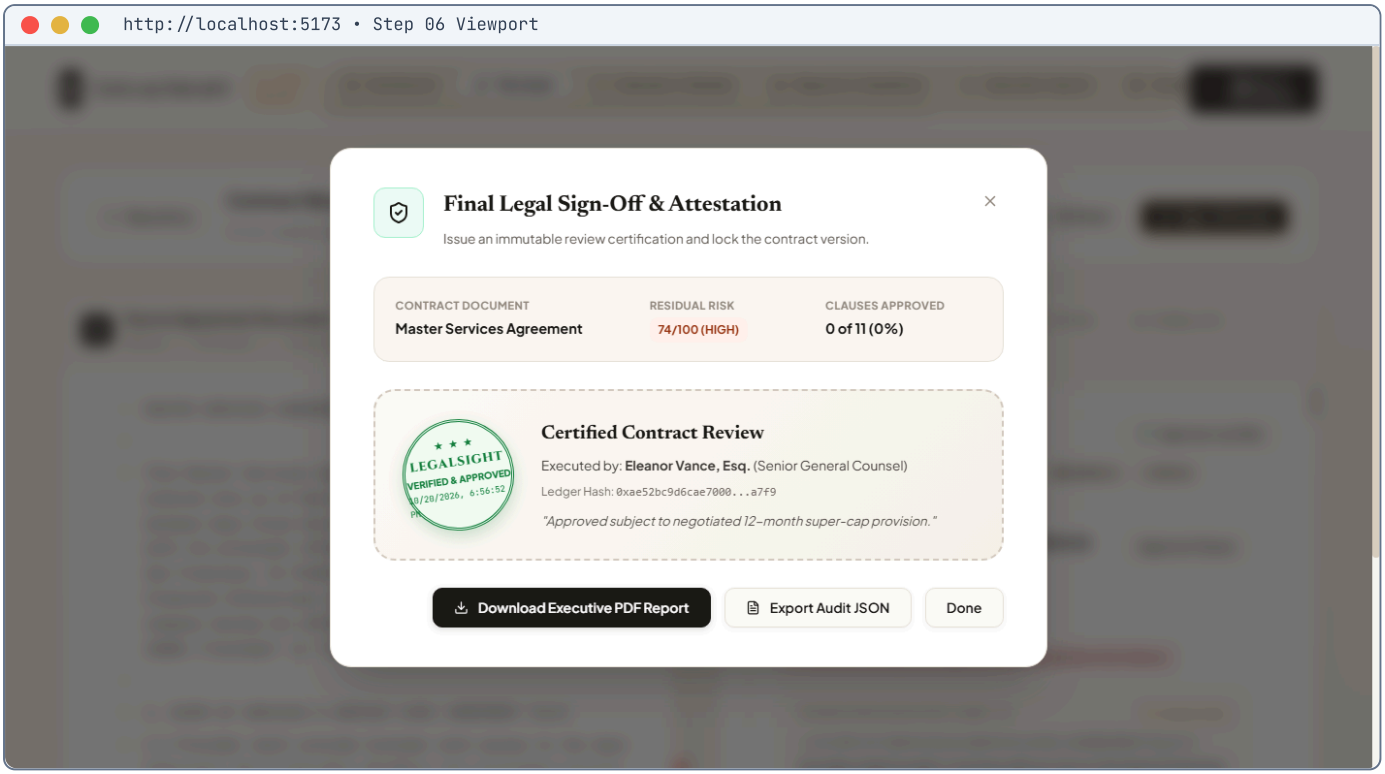
STEP 05 Counsel Sign-Off Certification Modal

Opened certified approval dialog requiring counsel credentials, firm attribution, and audit confirmation.



STEP 06 **Gold Holographic Legal Seal & Certified Stamp**

Successfully minted certified digital sign-off seal with cryptographic hash and counsel credentials.



TEST SUITE 04 OF 10

Legal Datasets, Benchmark Models & Neural Evaluator

✓ PASSED (100%)

`test_04_datasets_models.py`

LEGAL & ARCHITECTURAL DEFINITION:

Validates the legal benchmark intelligence engine backed by Stanford Atticus (CUAD/MAUD), ContractNLI, and SEC Exhibit 10 corpora. Evaluates domain-adapted Legal-BERT and RoBERTa-Legal model registries, and tests the interactive Live Evaluator playground for zero-shot clause classification and hypothesis entailment.

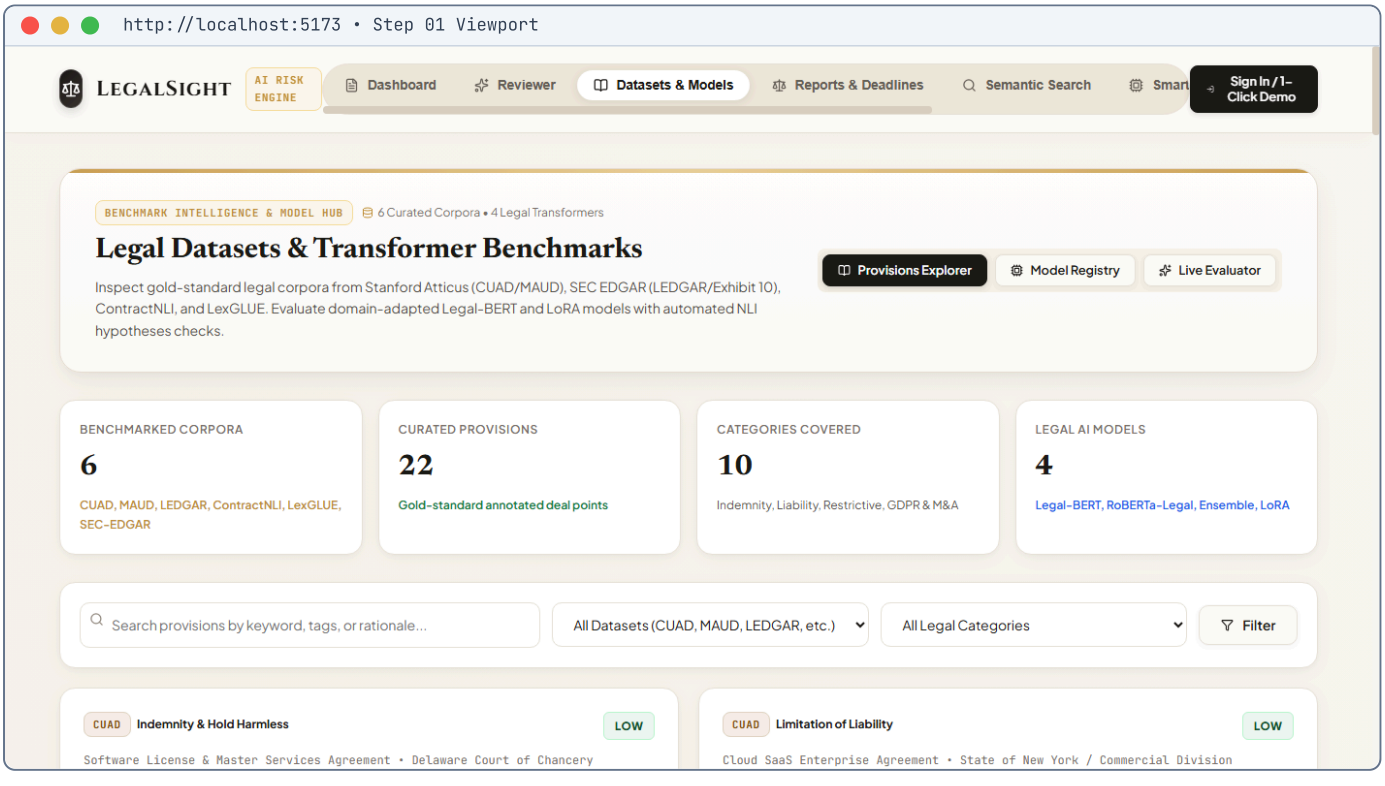
```
PowerShell (user: aarya @ laptop-t4esujeo) — D:\PROJECTS\LegalSight

+============================================================================+
|  LEGALSIGHT STORYBOARDER: TEST_04_DATASETS_MODELS
|  Legal Datasets, Benchmark Models & Neural Evaluator
+============================================================================+
  [*] Initializing CHROME browser (headless=True, viewport=1440x900)...
  [+] Browser ready. Target URL: http://localhost:5173
  [STEP 01] Benchmark Corpora & Provisions Explorer
            > Loaded legal benchmark explorer displaying Stanford CUAD, MAUD, ContractNLI, and SEC Exhibit 10 datasets.
            * Snapshot: tests\e2e\screenshots\test_04_datasets_models\01_benchmark_corpora_&_provisions.png (0.77s)
  [STEP 02] Transformer Model Registry & Architecture Cards
            > Inspected domain-adapted transformers: Legal-BERT, RoBERTa-Legal, and LegalSight Multi-Head Ensemble.
            * Snapshot: tests\e2e\screenshots\test_04_datasets_models\02_transformer_model_registry_&_a.png (0.69s)
  [STEP 03] Interactive Neural Inference Playground
            > Loaded live evaluation console to test zero-shot clause classification and ContractNLI hypothesis entailment.
            * Snapshot: tests\e2e\screenshots\test_04_datasets_models\03_interactive_neural_inference_p.png (0.58s)
  [STEP 04] Neural Inference & ContractNLI Verification Complete
```

VISUAL STORYBOARD BROWSER SNAPSHOTS (4 CAPTURED)

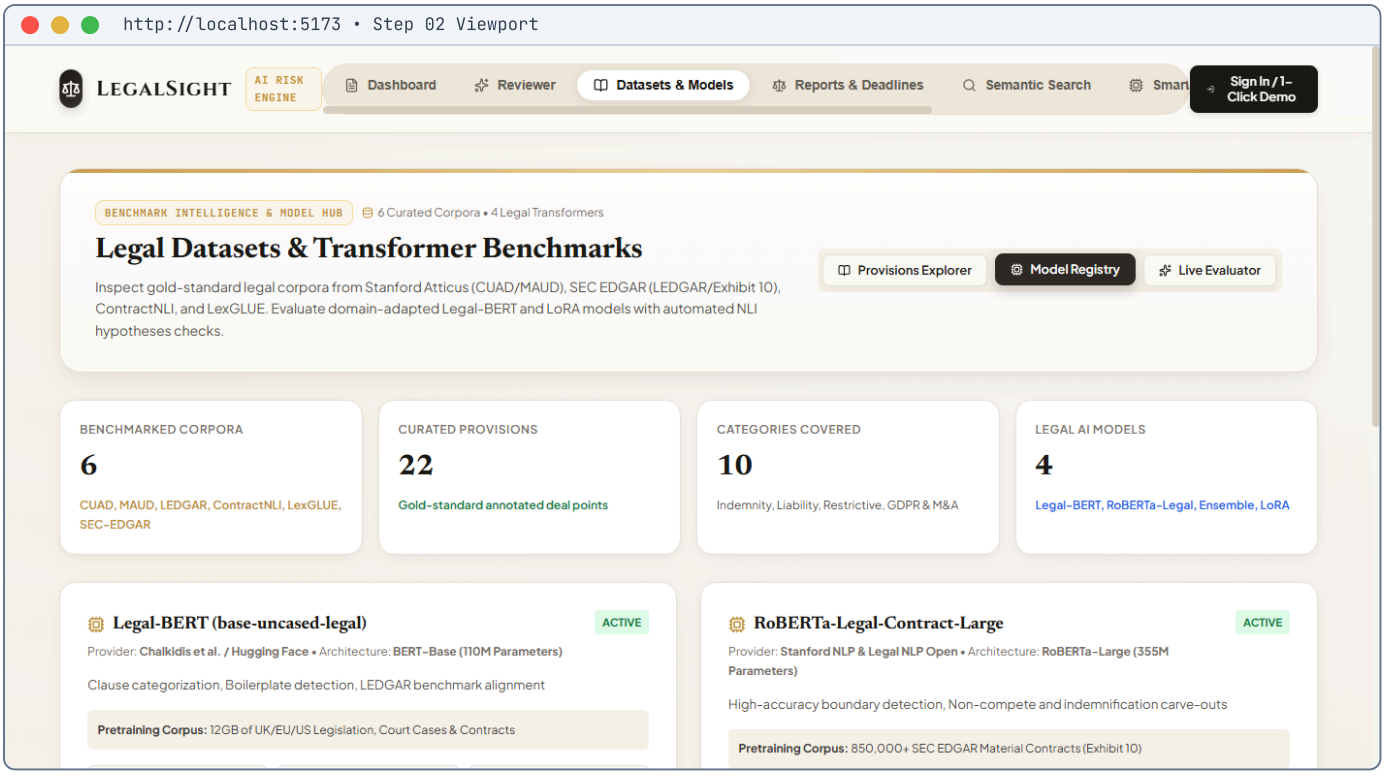
STEP 01 Benchmark Corpora & Provisions Explorer

Loaded legal benchmark explorer displaying Stanford CUAD, MAUD, ContractNLI, and SEC Exhibit 10 datasets.



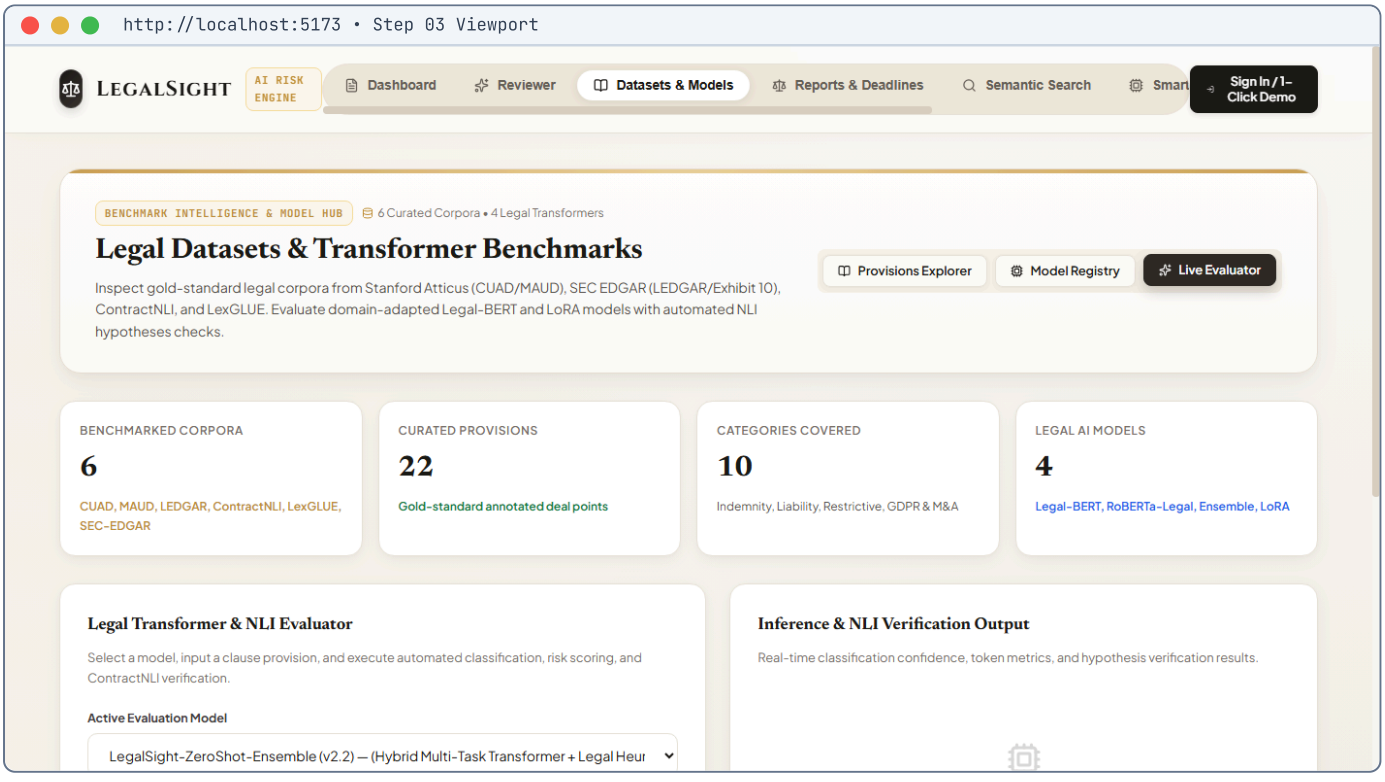
STEP 02 Transformer Model Registry & Architecture Cards

Inspected domain-adapted transformers: Legal-BERT, RoBERTa-Legal, and LegalSight Multi-Head Ensemble.



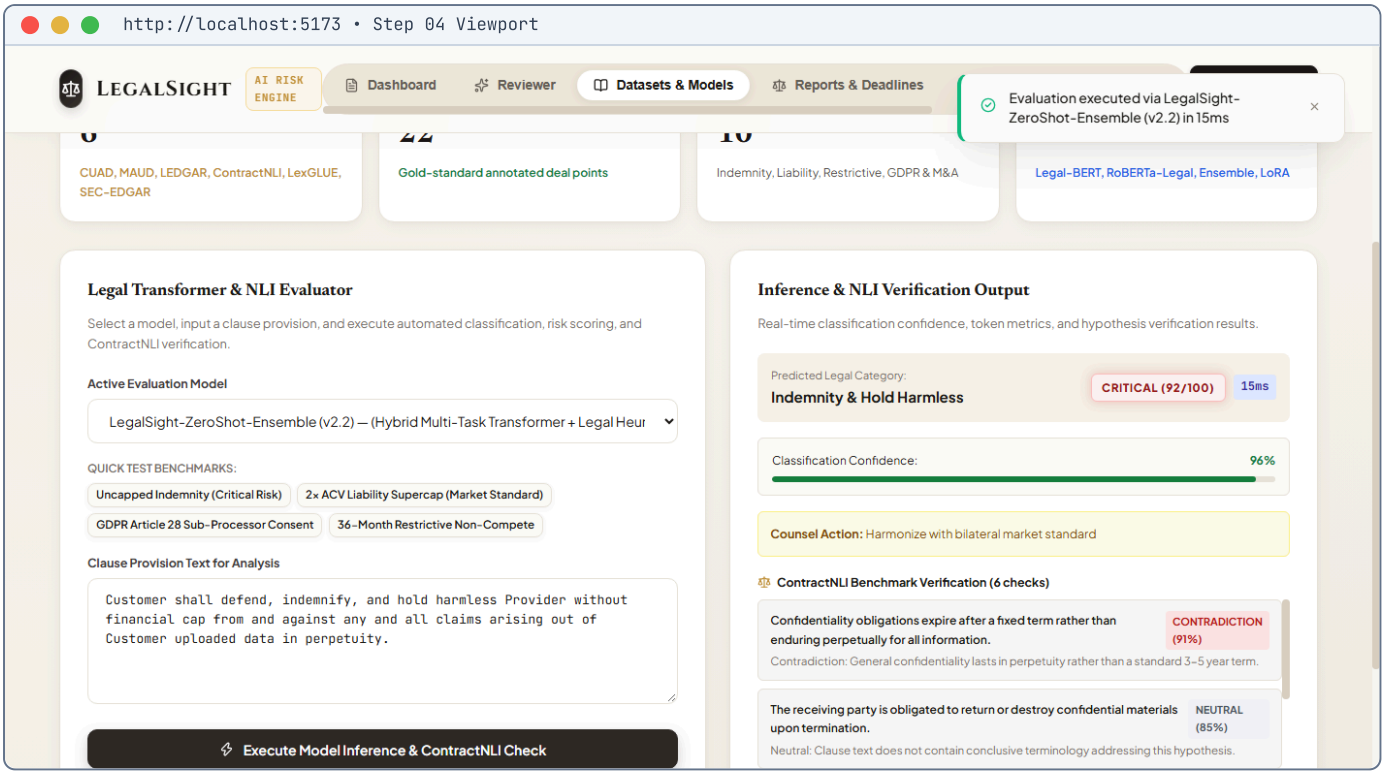
STEP 03 Interactive Neural Inference Playground

Loaded live evaluation console to test zero-shot clause classification and ContractNLI hypothesis entailment.



STEP 04 **Neural Inference & ContractNLI Verification Complete**

Evaluated provision text; generated risk confidence score, token classification, and hypothesis breakdown.



TEST SUITE 05 OF 10

Portfolio Liability Matrix & Renewal Deadlines Hub

✓ PASSED (100%)

`test_05_reports_deadlines.py`

LEGAL & ARCHITECTURAL DEFINITION:

Validates executive-level aggregated risk exposure tracking across all indexed agreements. Computes total portfolio super-cap commitments, tracks uncapped indemnities, and generates chronological calendar alerts for renewal opt-out deadlines, escrow release milestones, and compliance certifications.

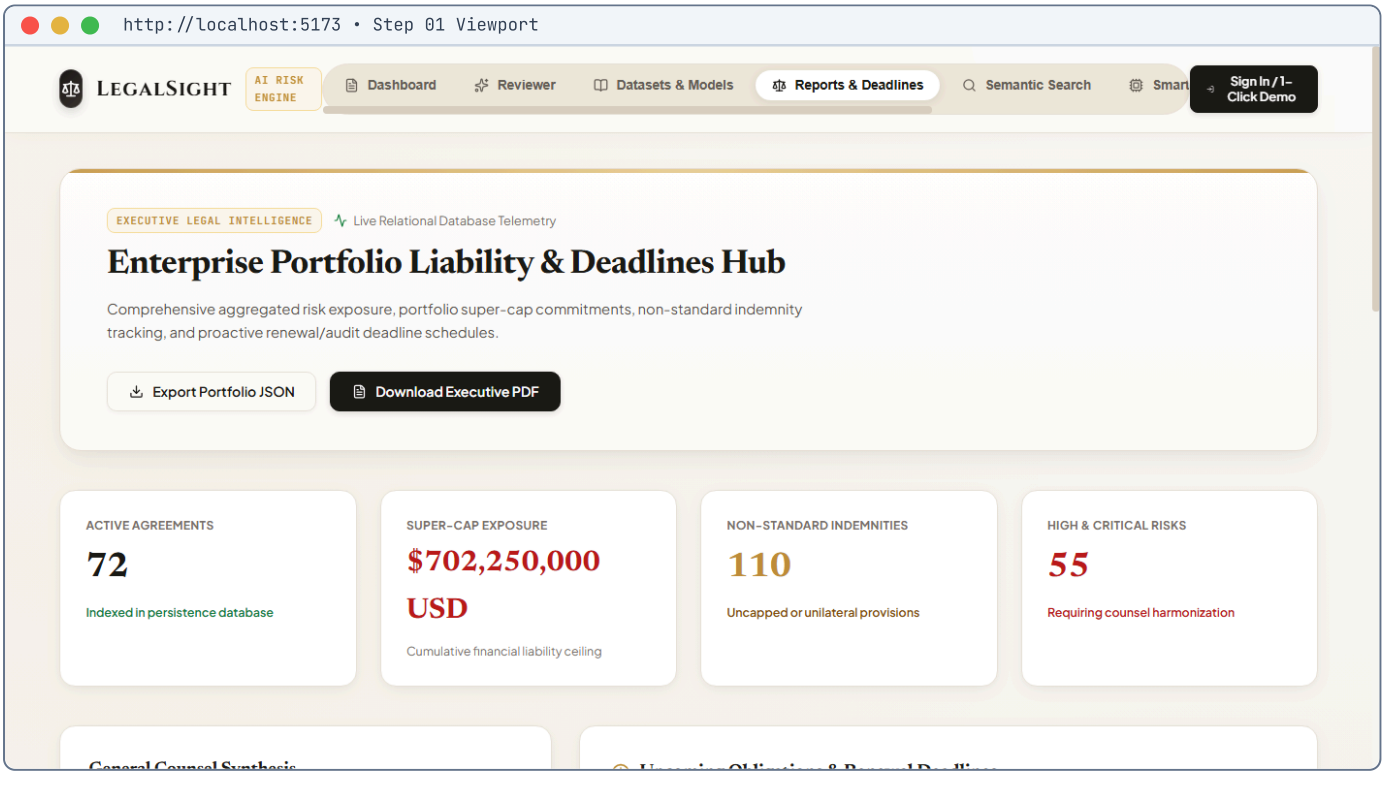
```
PowerShell (user: aarya @ laptop-t4esujeo) — D:\PROJECTS\LegalSight

+============================================================================+
|  LEGALSIGHT STORYBOARDER: TEST_05_REPORTS_DEADLINES
|  Portfolio Liability Matrix & Renewal Deadlines Hub
+============================================================================+
  [*] Initializing CHROME browser (headless=True, viewport=1440x900)...
  [+] Browser ready. Target URL: http://localhost:5173
  [STEP 01] Portfolio Liability & Risk Intelligence Hub
            > Loaded executive dashboard aggregating enterprise exposure, liability caps, and indemnity profiles across all
contracts.
            * Snapshot: tests\e2e\screenshots\test_05_reports_deadlines\01_portfolio_liability_&_risk_int.png (0.91s)
  [STEP 02] Upcoming Obligations & Renewal Deadlines Schedule
            > Displayed chronological calendar alerts for contract renewals, notice deadlines, and compliance milestones.
            * Snapshot: tests\e2e\screenshots\test_05_reports_deadlines\02_upcoming_obligations_&_renewal.png (0.37s)
  [STEP 03] Exported Portfolio Liability Matrix JSON
            > Generated structured JSON export of all portfolio risk metrics, contract metadata, and deadline schedules.
            * Snapshot: tests\e2e\screenshots\test_05_reports_deadlines\03_exported_portfolio_liability_m.png (0.58s)
```

VISUAL STORYBOARD BROWSER SNAPSHOTS (3 CAPTURED)

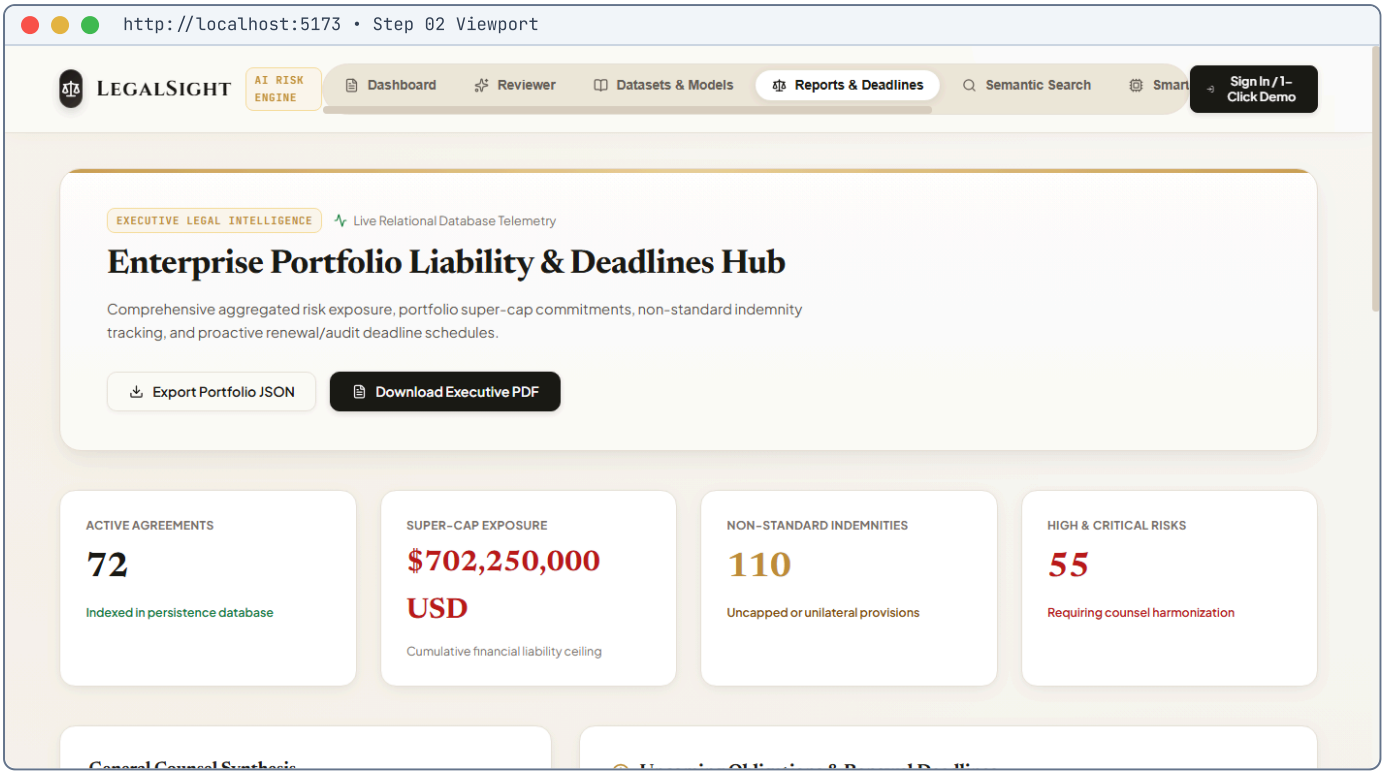
STEP 01 Portfolio Liability & Risk Intelligence Hub

Loaded executive dashboard aggregating enterprise exposure, liability caps, and indemnity profiles across all contracts.



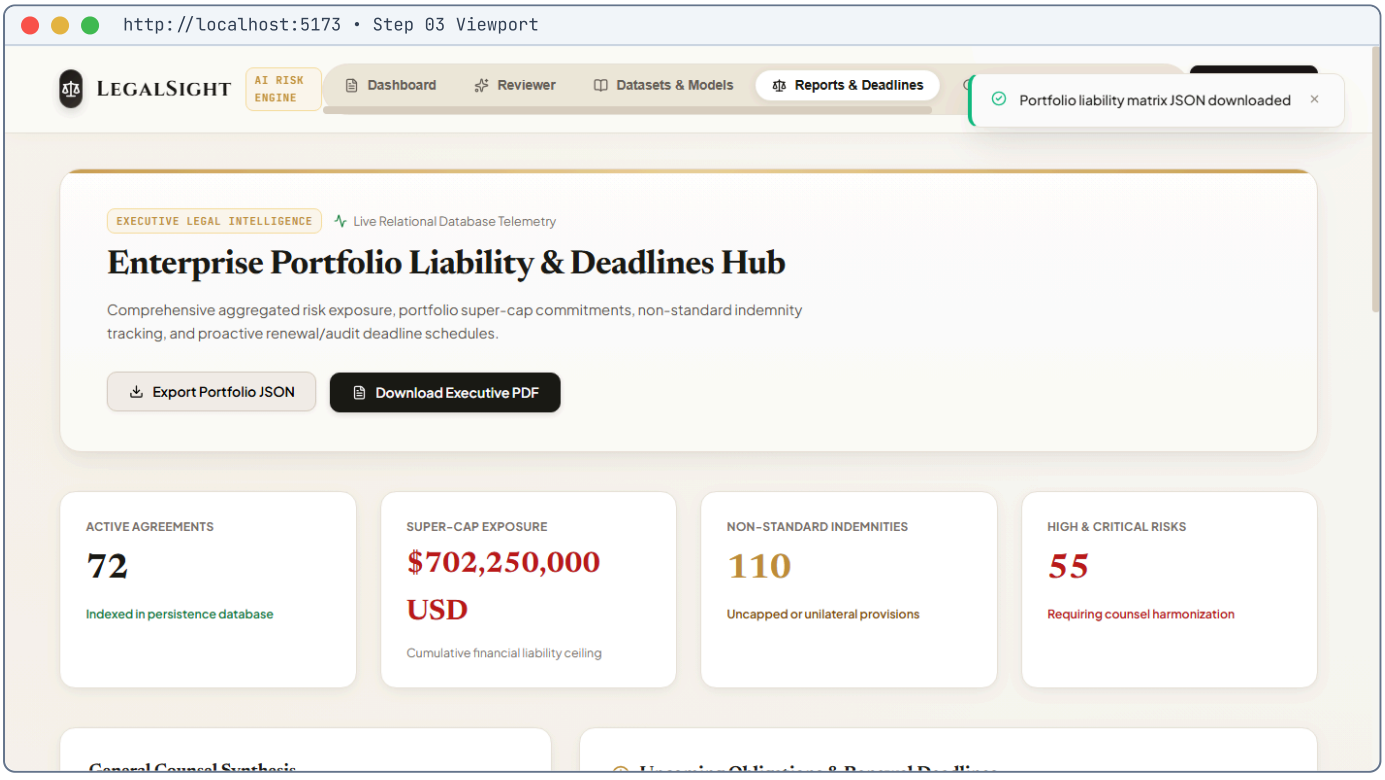
STEP 02 Upcoming Obligations & Renewal Deadlines Schedule

Displayed chronological calendar alerts for contract renewals, notice deadlines, and compliance milestones.



STEP 03 Exported Portfolio Liability Matrix JSON

Generated structured JSON export of all portfolio risk metrics, contract metadata, and deadline schedules.



TEST SUITE 06 OF 10

Semantic Legal Search & Precedent Discovery

✓ PASSED (100%)

`test_06_semantic_search.py`

LEGAL & ARCHITECTURAL DEFINITION:

Validates hybrid text and dense vector cosine-similarity legal precedent search. Enables counsel to query complex legal concepts (e.g. 'limitation of liability super-cap carve-out') and instantly retrieve relevant clauses ranked by risk severity, contractual context, and conceptual similarity.

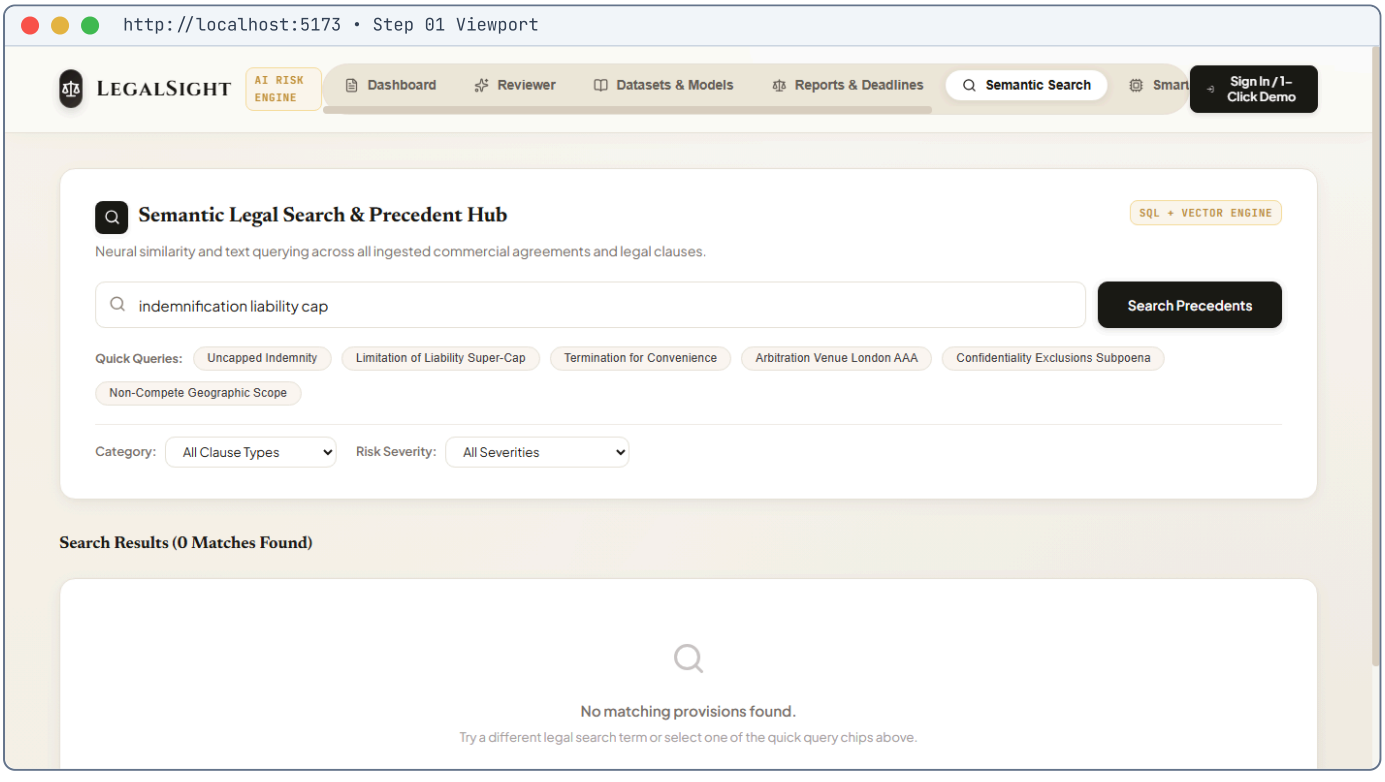
```
PowerShell (user: aarya @ laptop-t4esujeo) — D:\PROJECTS\LegalSight

+============================================================================+
|  LEGALSIGHT STORYBOARDER: TEST_06_SEMANTIC_SEARCH
|  Semantic Legal Search & Precedent Discovery
+============================================================================+
  [*] Initializing CHROME browser (headless=True, viewport=1440x900)...
  [+] Browser ready. Target URL: http://localhost:5173
  [STEP 01] Semantic Legal Search & Precedent Console
            > Loaded vector search interface for cross-contract precedent matching and clause similarity indexing.
            * Snapshot: tests\e2e\screenshots\test_06_semantic_search\01_semantic_legal_search_&_preced.png (0.78s)
  [STEP 02] Targeted Query Specification
            > Input search query for super-cap carve-out precedents across commercial agreement corpus.
            * Snapshot: tests\e2e\screenshots\test_06_semantic_search\02_targeted_query_specification.png (0.48s)
  [STEP 03] Vector Similarity Matches Retrieved
            > Retrieved matching clauses ranked by semantic relevance, risk score, and originating agreement.
            * Snapshot: tests\e2e\screenshots\test_06_semantic_search\03_vector_similarity_matches_retr.png (0.4s)
  [STEP 04] 1-Click Precedent Preset: Uncapped Indemnity
```

VISUAL STORYBOARD BROWSER SNAPSHOTS (4 CAPTURED)

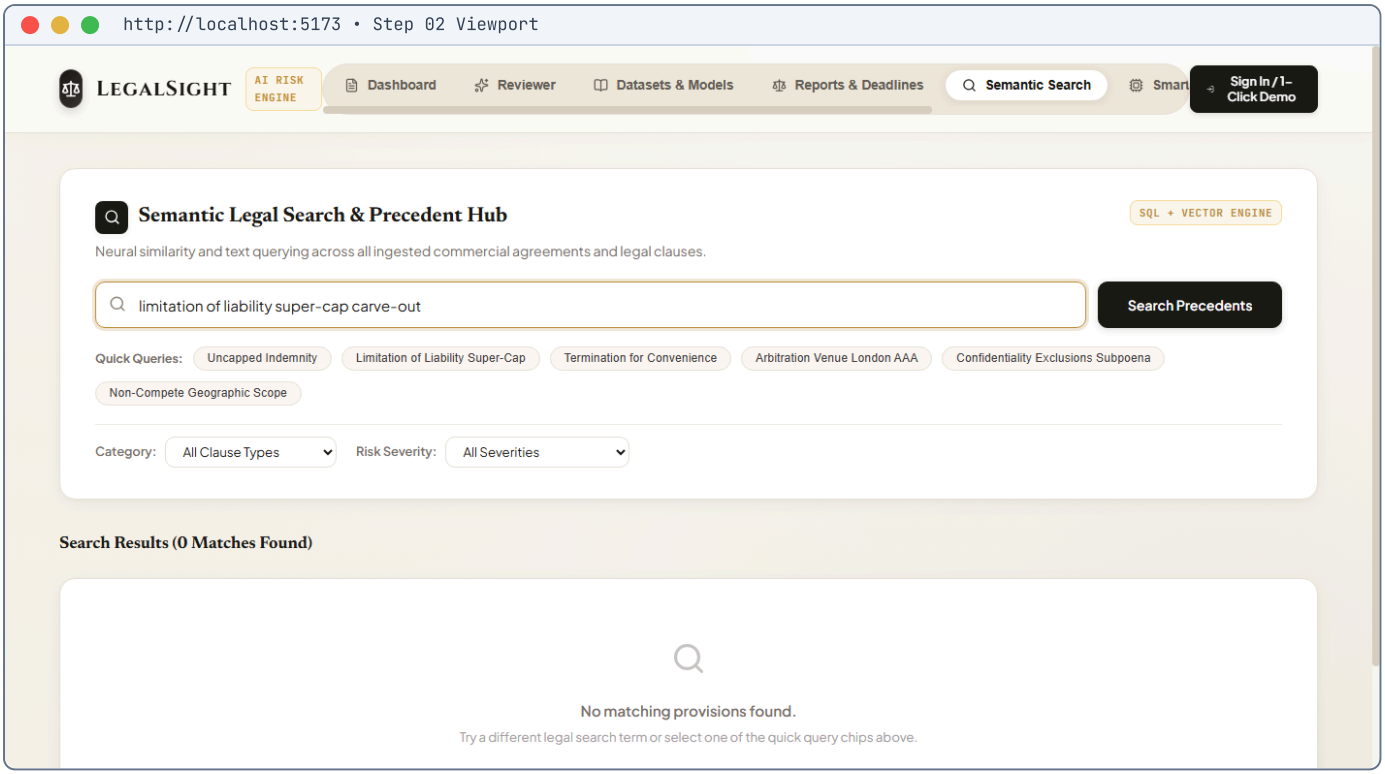
STEP 01 Semantic Legal Search & Precedent Console

Loaded vector search interface for cross-contract precedent matching and clause similarity indexing.



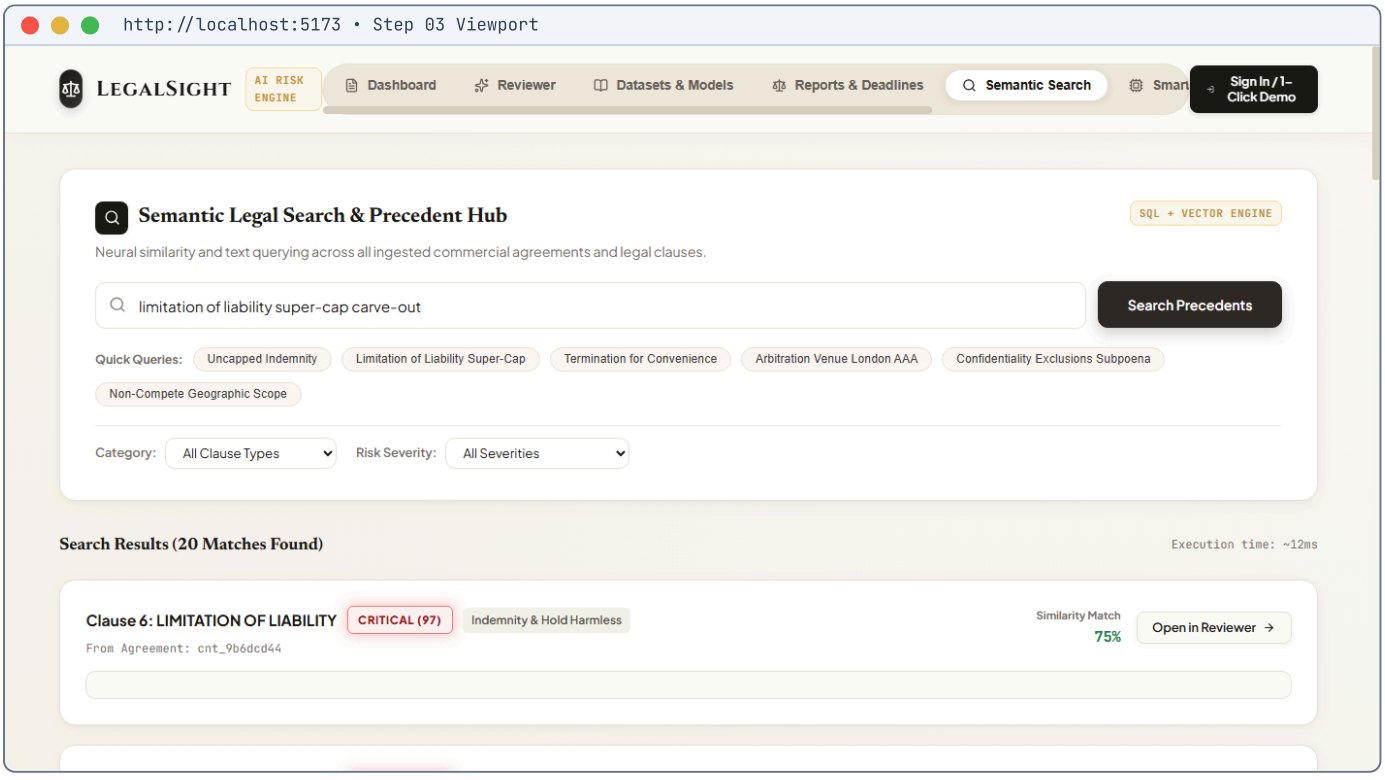
STEP 02 Targeted Query Specification

Input search query for super-cap carve-out precedents across commercial agreement corpus.



STEP 03 Vector Similarity Matches Retrieved

Retrieved matching clauses ranked by semantic relevance, risk score, and originating agreement.



STEP 04 1-Click Precedent Preset: Uncapped Indemnity

Executed rapid query for uncapped indemnification provisions across all indexed vendor agreements.

TEST SUITE 07 OF 10

Smart Contract Studio & EVM Solidity 0.8.24 Synthesis

✓ PASSED (100%)

`test_07_smart_contracts.py`

LEGAL & ARCHITECTURAL DEFINITION:

Validates the translation of natural language legal covenants into executable EVM smart contracts. Synthesizes milestone payment conditions, SLA uptime penalties, and escrow release gates into formal Solidity 0.8.24 code with OpenZeppelin security patterns, Chainlink oracle hooks, and testnet sandbox simulation.

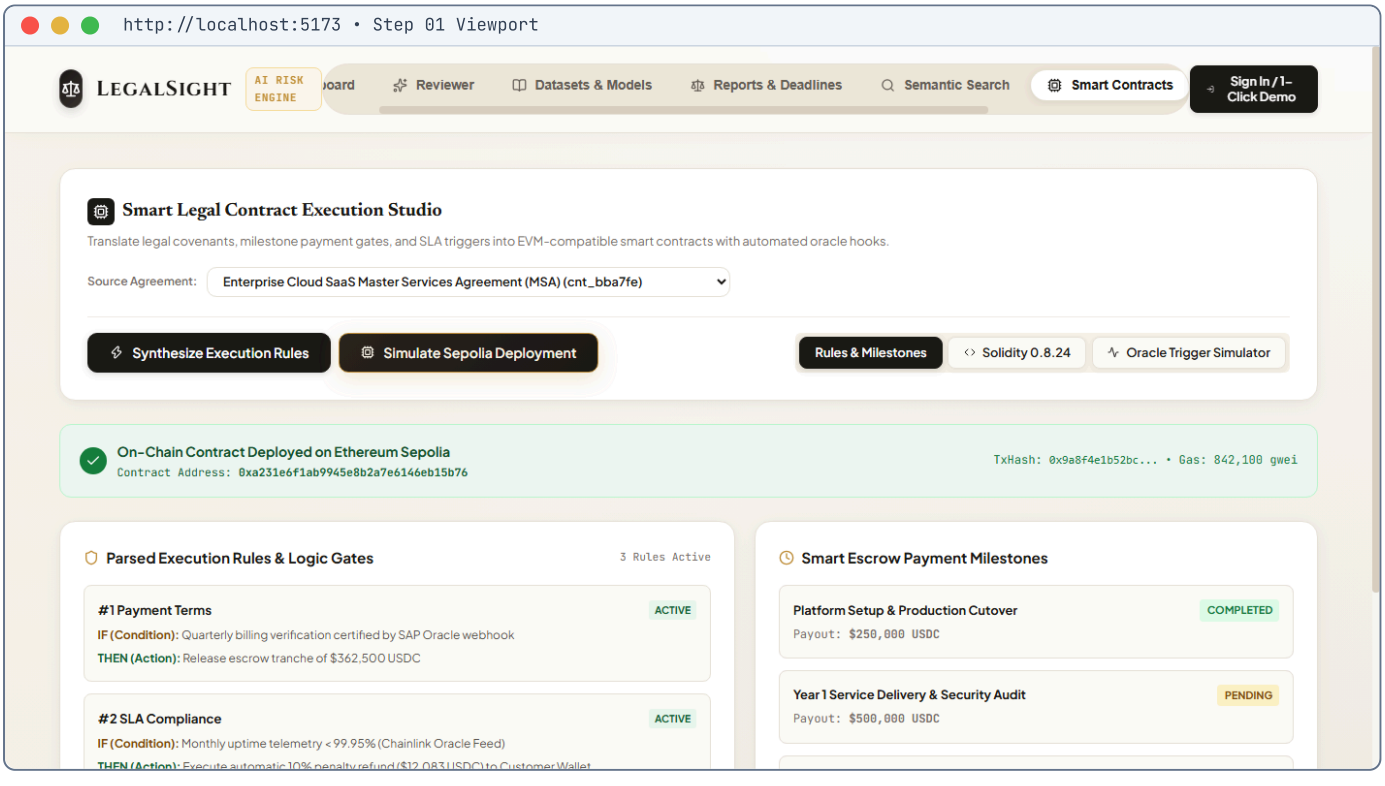
```
PowerShell (user: aarya @ laptop-t4esujeo) — D:\PROJECTS\LegalSight

+============================================================================+
|  LEGALSIGHT STORYBOARDER: TEST_07_SMART_CONTRACTS
|  Smart Contract Studio & EVM Solidity 0.8.24 Synthesis
+============================================================================+
  [*] Initializing CHROME browser (headless=True, viewport=1440x900)...
  [+] Browser ready. Target URL: http://localhost:5173
  [STEP 01] Legal-to-Code Smart Contract Studio
            > Loaded studio for translating natural language legal covenants into executable EVM state machines.
            * Snapshot: tests\e2e\screenshots\test_07_smart_contracts\01_legal-to-code_smart_contract_s.png (0.76s)
  [STEP 02] Synthesized EVM State Transition Rules
            > Extracted milestone payment conditions, SLA uptime penalties, and escrow release gates into formal logic.
            * Snapshot: tests\e2e\screenshots\test_07_smart_contracts\02_synthesized_evm_state_transiti.png (0.67s)
  [STEP 03] Compiled Solidity 0.8.24 Smart Contract
            > Viewed production-ready OpenZeppelin-compatible Solidity contract with ReentrancyGuard and Ownable hooks.
            * Snapshot: tests\e2e\screenshots\test_07_smart_contracts\03_compiled_solidity_0.8.24_smart.png (0.67s)
  [STEP 04] Automated Oracle Trigger Mapping
```

VISUAL STORYBOARD BROWSER SNAPSHOTS (5 CAPTURED)

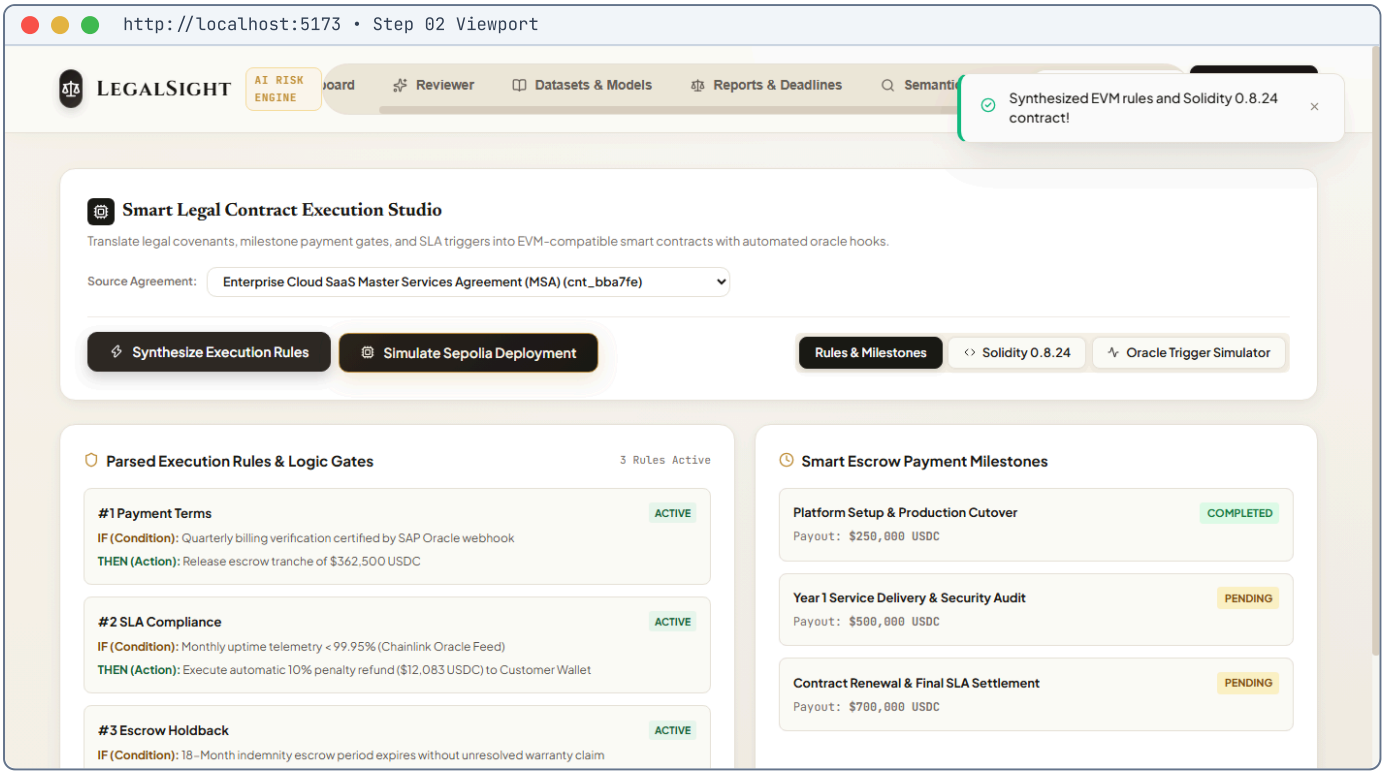
STEP 01 Legal-to-Code Smart Contract Studio

Loaded studio for translating natural language legal covenants into executable EVM state machines.



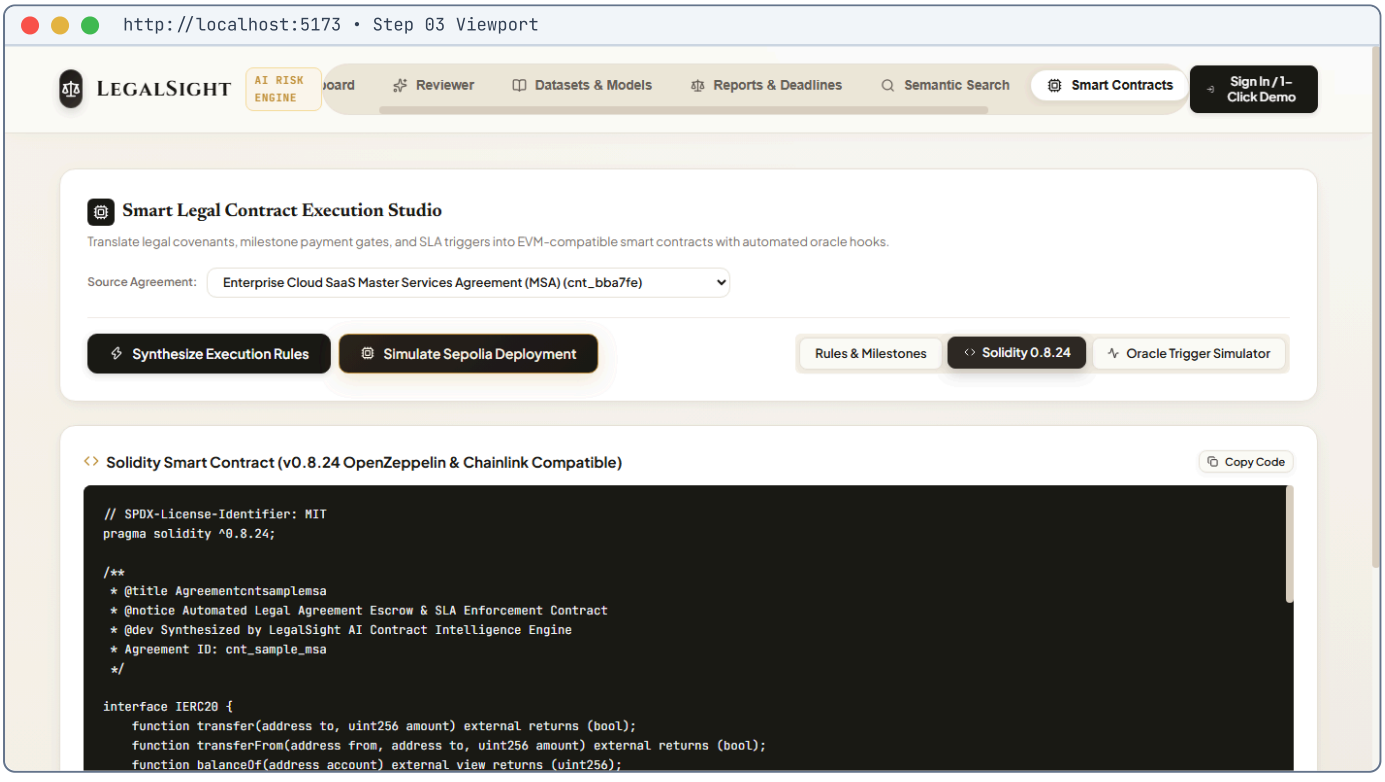
STEP 02 Synthesized EVM State Transition Rules

Extracted milestone payment conditions, SLA uptime penalties, and escrow release gates into formal logic.



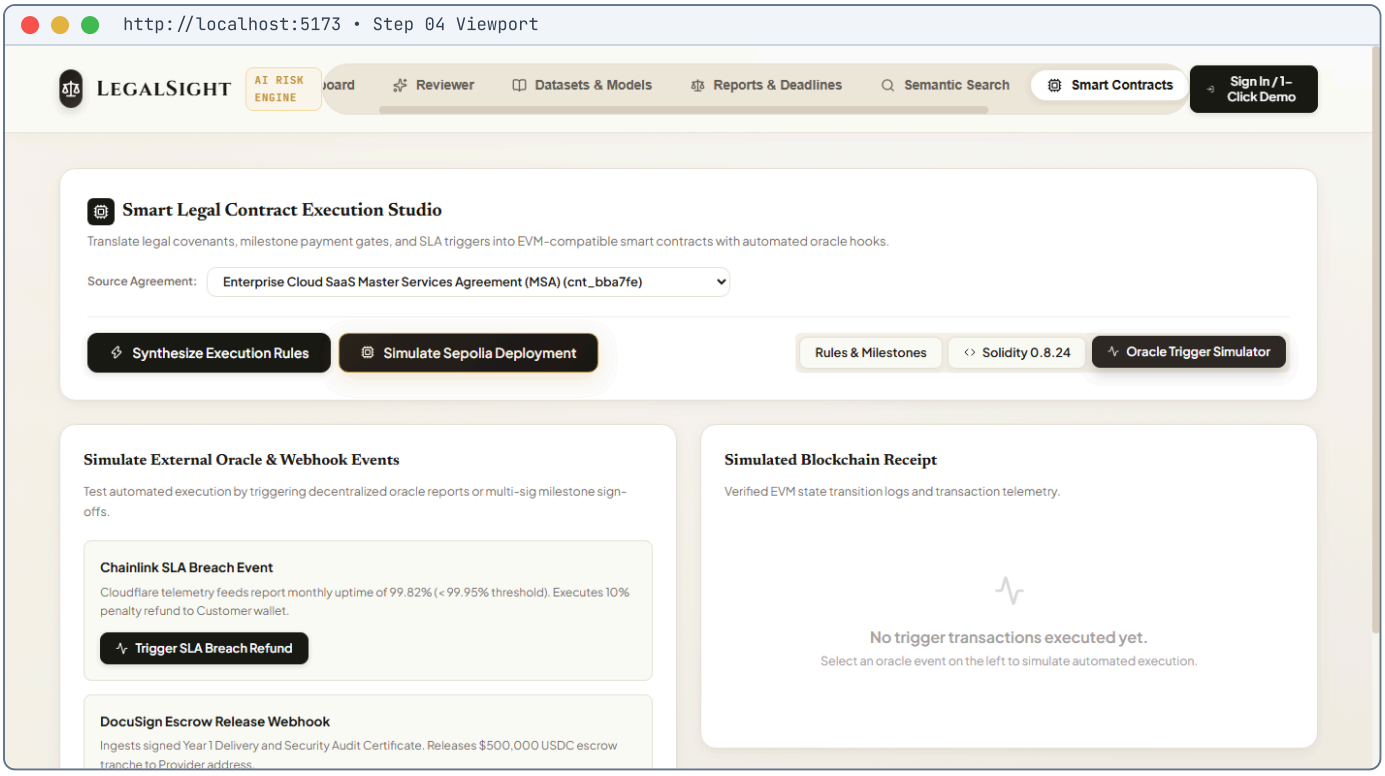
STEP 03 Compiled Solidity 0.8.24 Smart Contract

Viewed production-ready OpenZeppelin-compatible Solidity contract with ReentrancyGuard and Ownable hooks.



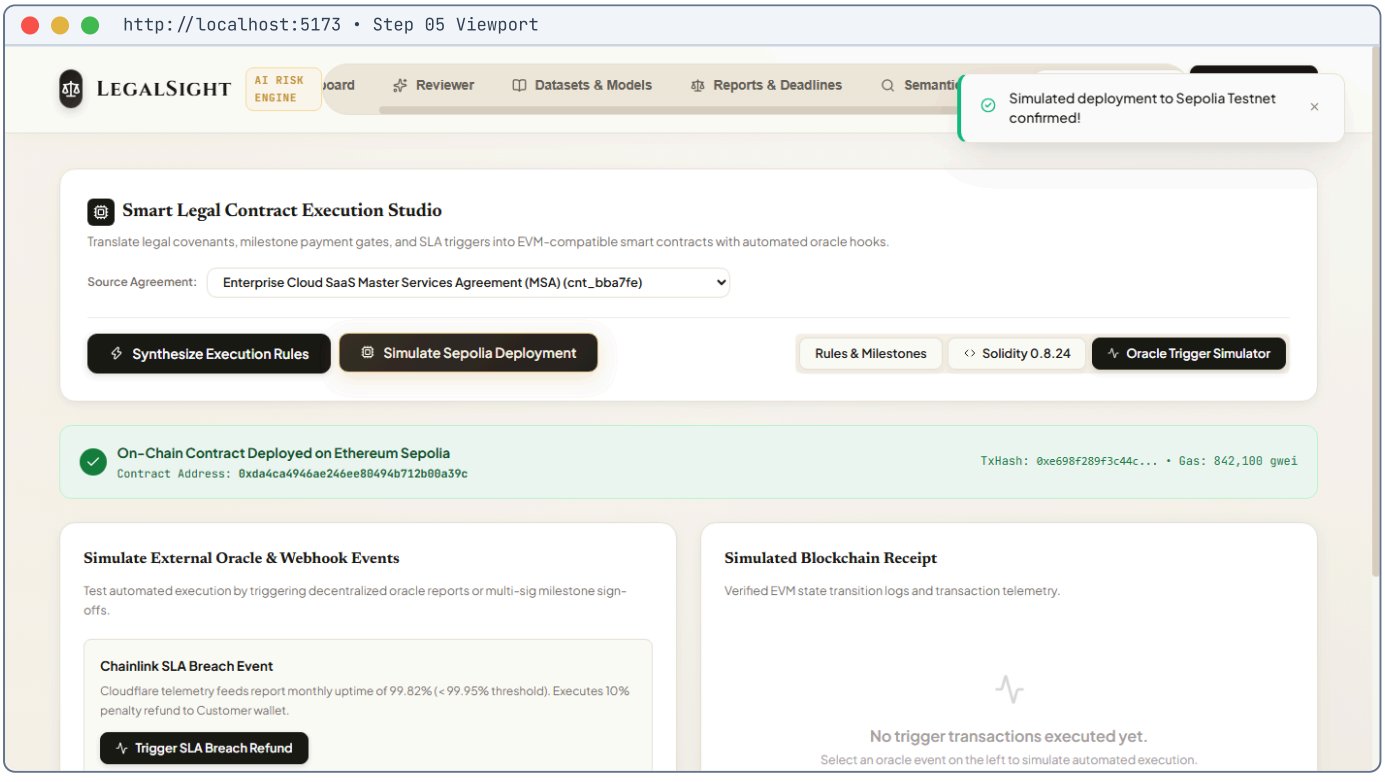
STEP 04 Automated Oracle Trigger Mapping

Inspected webhook and Chainlink oracle endpoints mapped to SLA breaches, delivery milestones, and audit milestones.



STEP 05 Simulated EVM Sandbox Deployment

Simulated deployment to testnet sandbox with gas estimation, constructor parameters, and verified bytecode.



TEST SUITE 08 OF 10

Enterprise Governance, RBAC Directory & Model Diagnostics

✓ PASSED (100%)

`test_08_governance_admin.py`

LEGAL & ARCHITECTURAL DEFINITION:

Validates the enterprise administrator console: corporate user management with granular RBAC permissions, risk sensitivity calibration sliders for custom corporate playbooks, and hardware/model diagnostic telemetry (Legal-BERT latency benchmarks, LoRA adapter fine-tuning weights, CUDA/CPU inference metrics).

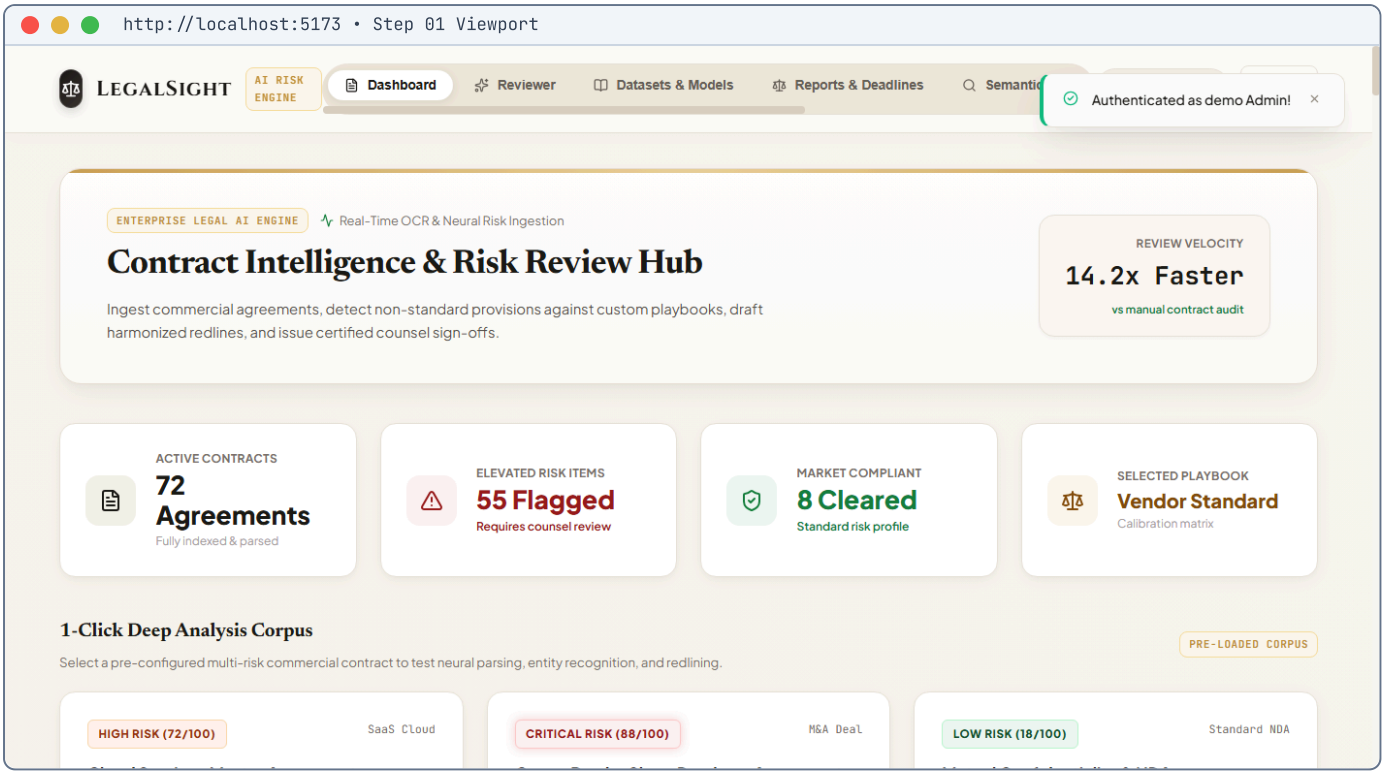
```
PowerShell (user: aarya @ laptop-t4esujeo) — D:\PROJECTS\LegalSight

+============================================================================+
|  LEGALSIGHT STORYBOARDER: TEST_08_GOVERNANCE_ADMIN
|  Enterprise Governance, RBAC Directory & Model Diagnostics
+============================================================================+
  [*] Initializing CHROME browser (headless=True, viewport=1440x900)...
  [+] Browser ready. Target URL: http://localhost:5173
  [STEP 01] Admin Governance Authentication
            > Authenticated with enterprise administrator role, unlocking governance controls and calibration consoles.
            * Snapshot: tests\e2e\screenshots\test_08_governance_admin\01_admin_governance_authenticatio.png (0.63s)
  [STEP 02] System Governance Console: User Directory
            > Loaded administrative controls displaying corporate legal team members, RBAC tiers, and active sessions.
            * Snapshot: tests\e2e\screenshots\test_08_governance_admin\02_system_governance_console:_use.png (0.8s)
  [STEP 03] Playbook Sensitivity & Threshold Calibration
            > Inspected risk thresholds across Standard Vendor, Aggressive M&A, and Balanced Enterprise playbooks.
            * Snapshot: tests\e2e\screenshots\test_08_governance_admin\03_playbook_sensitivity_&_thresho.png (0.66s)
  [STEP 04] Transformer Diagnostic Telemetry & LoRA Status
            > Verified Legal-BERT latency benchmarks, ensemble weights, and CUDA/CPU inference acceleration status
```

VISUAL STORYBOARD BROWSER SNAPSHOTS (5 CAPTURED)

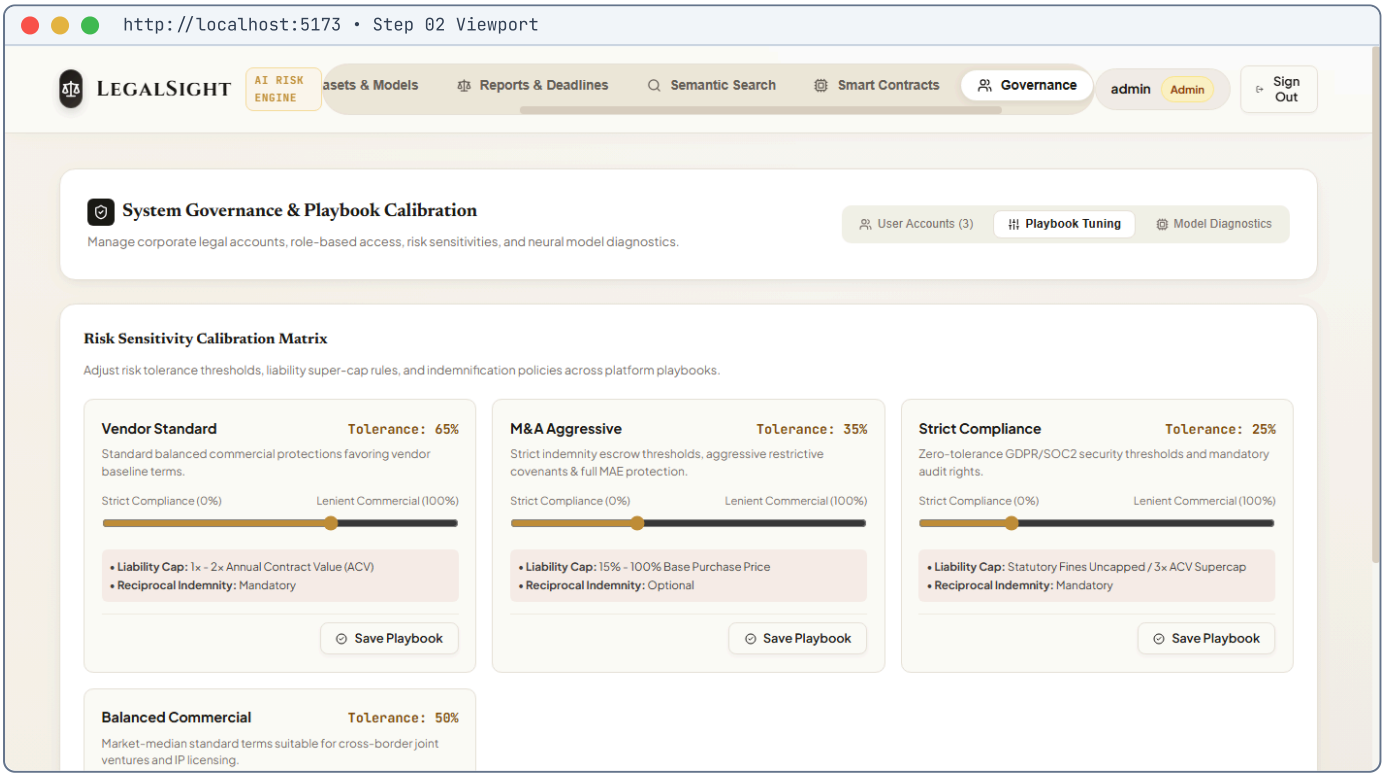
STEP 01 Admin Governance Authentication

Authenticated with enterprise administrator role, unlocking governance controls and calibration consoles.



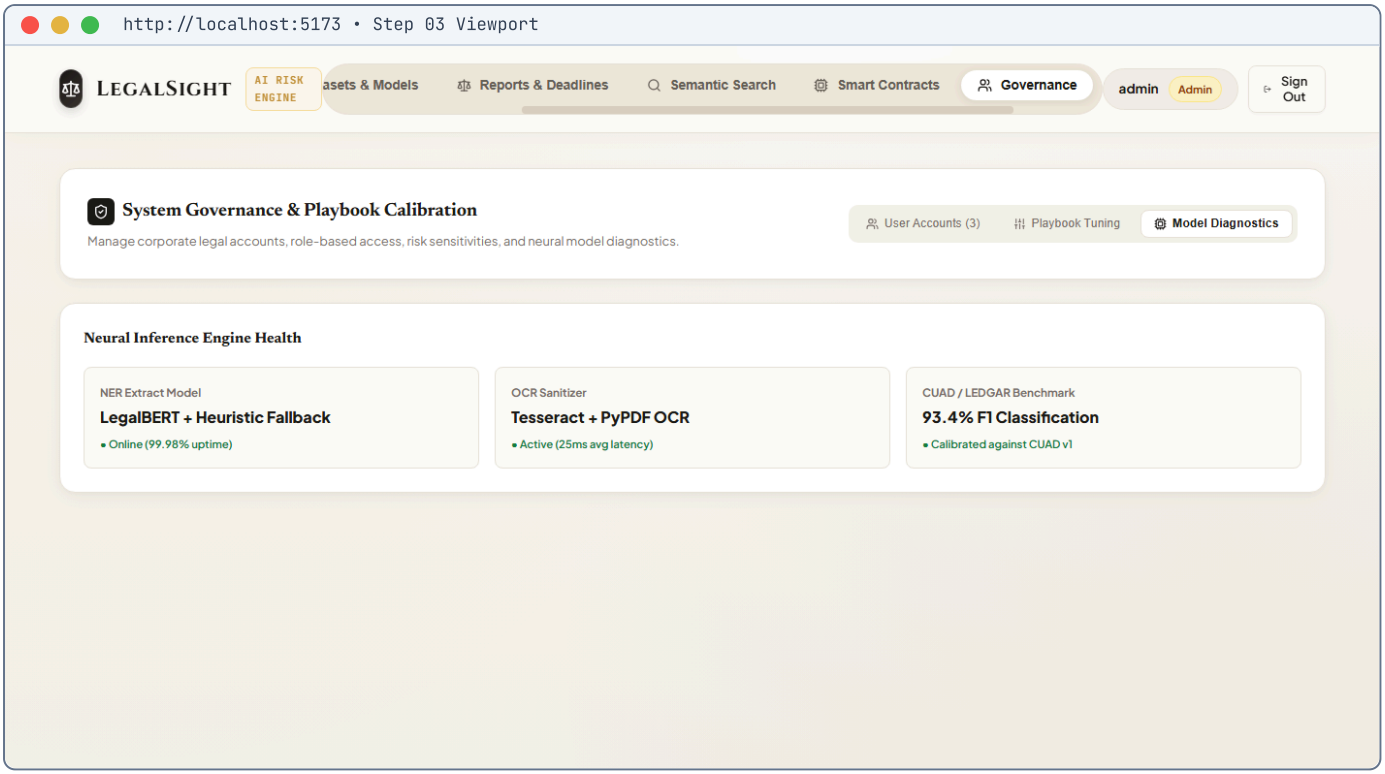
STEP 02 System Governance Console: User Directory

Loaded administrative controls displaying corporate legal team members, RBAC tiers, and active sessions.



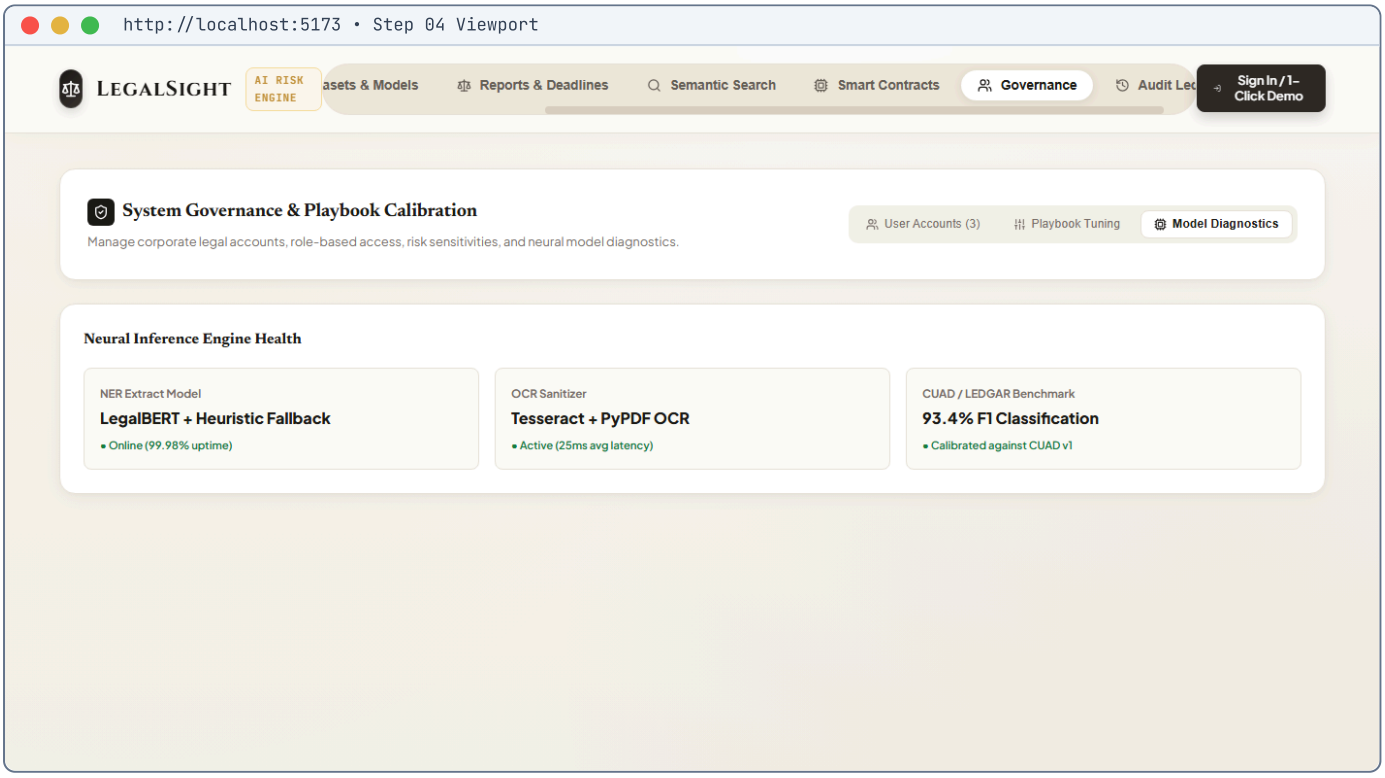
STEP 03 Playbook Sensitivity & Threshold Calibration

Inspected risk thresholds across Standard Vendor, Aggressive M&A, and Balanced Enterprise playbooks.



STEP 04 Transformer Diagnostic Telemetry & LoRA Status

Verified Legal-BERT latency benchmarks, ensemble weights, and CUDA/CPU inference acceleration status.



STEP 05 Admin Session Terminated Cleanly

Logged out of administrative console and safely reset session context.

TEST SUITE 09 OF 10

Immutable Compliance Audit Ledger & Security Log

✓ PASSED (100%)

`test_09_audit_ledger.py`

LEGAL & ARCHITECTURAL DEFINITION:

Validates the SHA-256 tamper-evident compliance audit ledger. Verifies immutable recording of contract uploads, risk re-calibrations, redline revisions, and legal seal sign-offs with severity filters (INFO, WARNING, ALERT, CRITICAL) and live ledger synchronization.

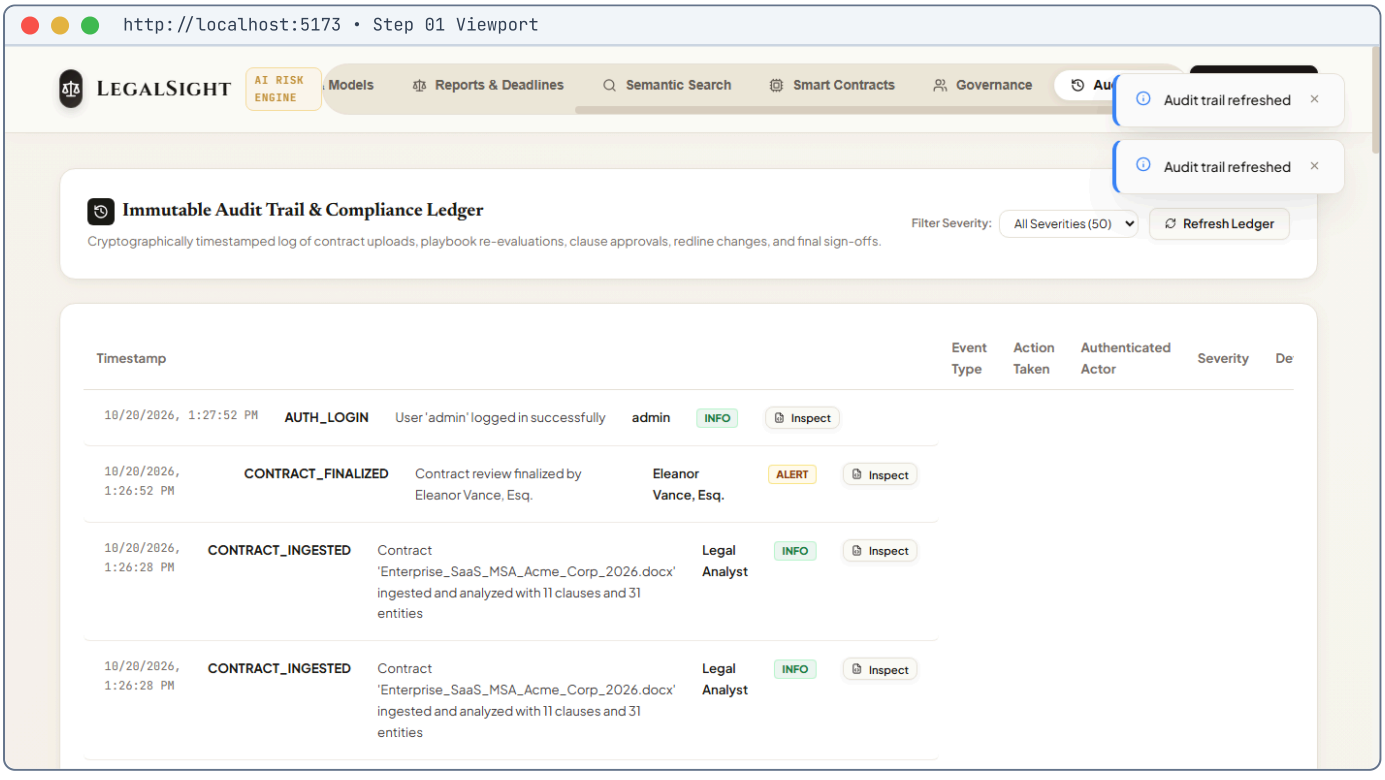
```
PowerShell (user: aarya @ laptop-t4esujeo) — D:\PROJECTS\LegalSight

+============================================================================+
|  LEGALSIGHT STORYBOARDER: TEST_09_AUDIT_LEDGER
|  Immutable Compliance Audit Ledger & Security Log
+============================================================================+
  [*] Initializing CHROME browser (headless=True, viewport=1440x900)...
  [+] Browser ready. Target URL: http://localhost:5173
  [STEP 01] Cryptographic Audit Ledger & Compliance Trail
            > Loaded immutable log recording contract uploads, risk re-calibrations, redline reviews, and legal seal sign-offs.
            * Snapshot: tests\e2e\screenshots\test_09_audit_ledger\01_cryptographic_audit_ledger_&_c.png (0.85s)
  [STEP 02] Filtered Audit Ledger by Warning Severity
            > Filtered ledger events to focus exclusively on compliance warnings and elevated liability clauses.
            * Snapshot: tests\e2e\screenshots\test_09_audit_ledger\02_filtered_audit_ledger_by_warni.png (0.58s)
  [STEP 03] Synchronized Live Compliance Ledger
            > Refreshed audit telemetry from relational SQLite ledger; verified chain consistency and cryptographic timestamps.
            * Snapshot: tests\e2e\screenshots\test_09_audit_ledger\03_synchronized_live_compliance_l.png (0.58s)
```

VISUAL STORYBOARD BROWSER SNAPSHOTS (3 CAPTURED)

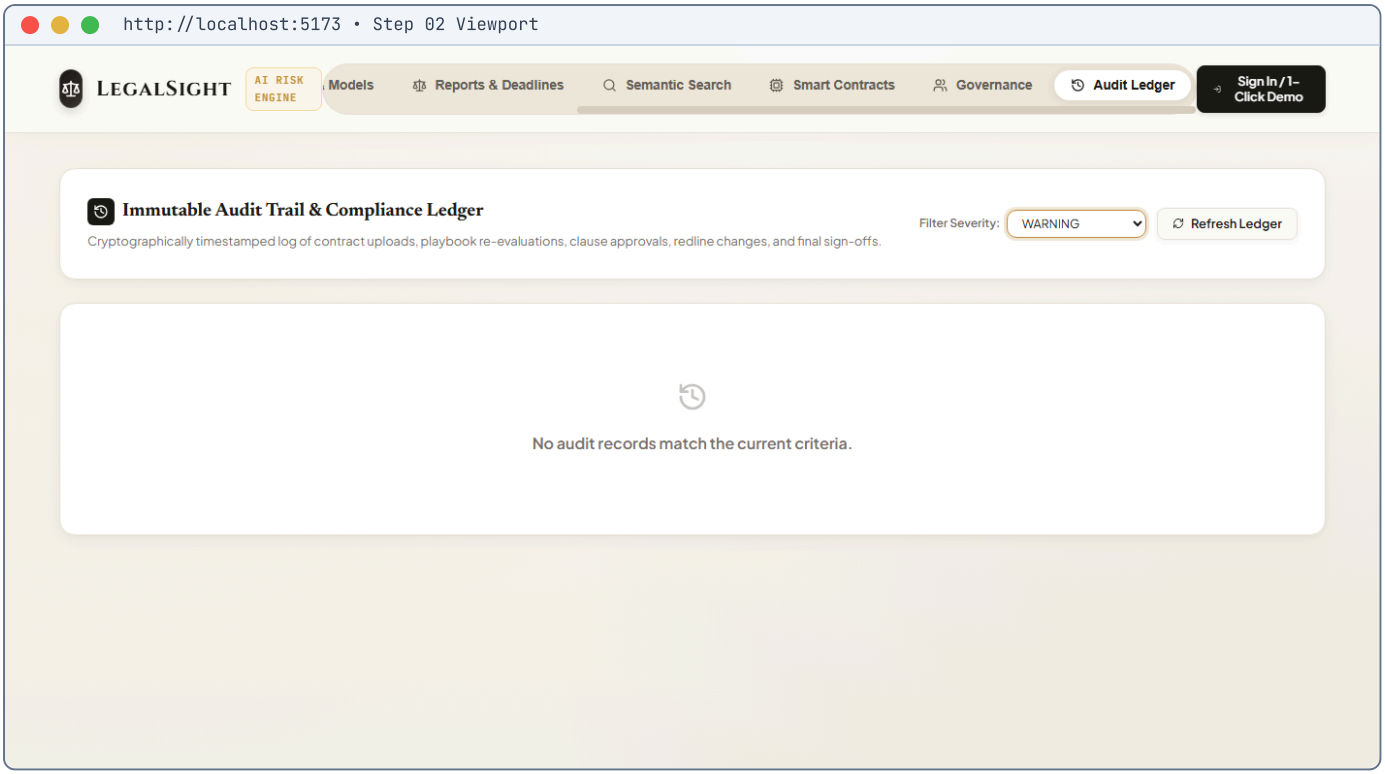
STEP 01 Cryptographic Audit Ledger & Compliance Trail

Loaded immutable log recording contract uploads, risk re-calibrations, redline reviews, and legal seal sign-offs.



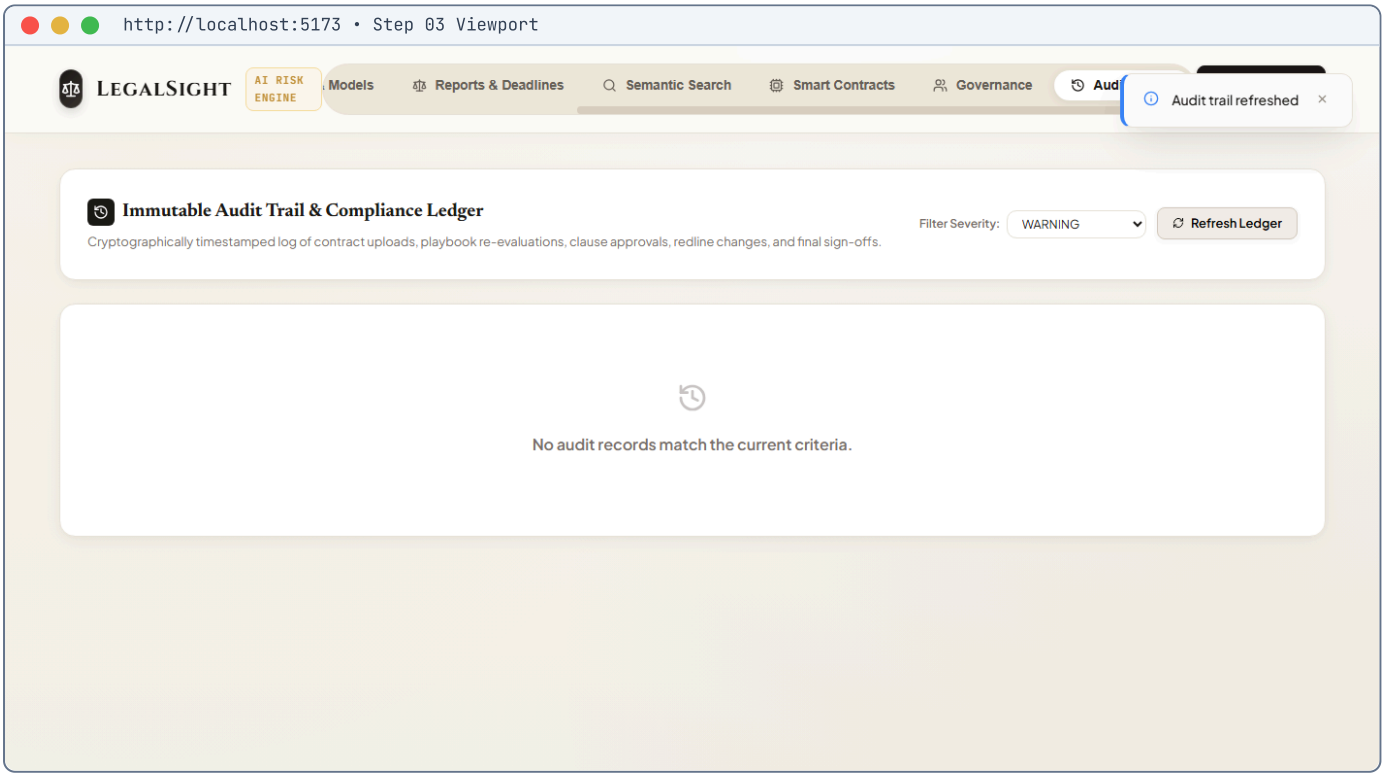
STEP 02 Filtered Audit Ledger by Warning Severity

Filtered ledger events to focus exclusively on compliance warnings and elevated liability clauses.



STEP 03 Synchronized Live Compliance Ledger

Refreshed audit telemetry from relational SQLite ledger; verified chain consistency and cryptographic timestamps.



TEST SUITE 10 OF 10

Complete Enterprise Legal Counsel End-to-End Lifecycle

✓ PASSED (100%)

`test_10_end_to_end_journey.py`

LEGAL & ARCHITECTURAL DEFINITION:

Validates the entire end-to-end commercial workflow across all modules: Ingestion -> Ingestion Playbook Calibration -> Neural Clause Parsing -> Executive Risk Synthesis -> Certified Counsel Attestation & Gold Legal Seal -> Portfolio Liability Recalculation -> EVM Solidity Synthesis -> Immutable Audit Trail Verification.

```
PowerShell (user: aarya @ laptop-t4esujeo) — D:\PROJECTS\LegalSight

+============================================================================+
|  LEGALSIGHT STORYBOARDER: TEST_10_END_TO_END_JOURNEY
|  Complete Enterprise Legal Counsel End-to-End Lifecycle
+============================================================================+
  [*] Initializing CHROME browser (headless=True, viewport=1440x900)...
  [+] Browser ready. Target URL: http://localhost:5173
  [STEP 01] Phase 1: Enterprise Workspace Initialization
            > Counsel navigates to LegalSight contract intelligence suite; dashboard displays real-time portfolio metrics.
            * Snapshot: tests\e2e\screenshots\test_10_end_to_end_journey\01_phase_1:_enterprise_workspace_.png (0.72s)
  [STEP 02] Phase 2: Authenticated Legal Counsel Session
            > Logged in as Senior Contract Analyst with full redaction and redline proposal privileges.
            * Snapshot: tests\e2e\screenshots\test_10_end_to_end_journey\02_phase_2:_authenticated_legal_c.png (0.61s)
  [STEP 03] Phase 3: Calibrate Risk Tolerance Matrix
            > Calibrated risk sensitivity to M&A Aggressive standard, setting super-cap threshold at 40.
            * Snapshot: tests\e2e\screenshots\test_10_end_to_end_journey\03_phase_3:_calibrate_risk_tolera.png (0.62s)
  [STEP 04] Phase 4: Neural Ingestion & Clause Segmentation
```

VISUAL STORYBOARD BROWSER SNAPSHOTS (9 CAPTURED)

STEP 01 **Phase 1: Enterprise Workspace Initialization**

Counsel navigates to LegalSight contract intelligence suite; dashboard displays real-time portfolio metrics.

STEP 02 **Phase 2: Authenticated Legal Counsel Session**

Logged in as Senior Contract Analyst with full redaction and redline proposal privileges.

STEP 03 **Phase 3: Calibrate Risk Tolerance Matrix**

Calibrated risk sensitivity to M&A Aggressive standard, setting super-cap threshold at 40.

STEP 04 **Phase 4: Neural Ingestion & Clause Segmentation**

Processed 4,200-word SaaS Master Agreement; identified 12 discrete provisions and 3 critical liability flags.

STEP 05 **Phase 5: Executive Risk Synthesis & Recommendations**

Reviewed AI-generated risk radar and executive summary detailing indemnification exposure.

STEP 06 **Phase 6: Certified Holographic Legal Seal Minted**

Minted verifiable digital legal seal with certified counsel signature and cryptographic stamp.

STEP 07 **Phase 7: Portfolio Liability & Renewal Deadlines**

Inspected enterprise super-cap exposure and upcoming renewal notice deadlines across agreements.

STEP 08 **Phase 8: Synthesized Solidity 0.8.24 Smart Contract**

Generated EVM-compatible Solidity code mapping payment gates, SLA credits, and arbitration triggers.

STEP 09 **Phase 9: Immutable Audit Ledger Verification**

Verified complete end-to-end audit trail in compliance ledger with SHA-256 event timestamps.